Mathematical Simulation of Blood Flow

A Literature Review and Idealized Stenosis Study

Daniel Henderson

Michigan Technological University

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Abstract.

This report develops a mathematical and numerical framework for idealized stenotic blood-flow simulation, connecting incompressible Navier–Stokes and generalized-Newtonian continuum models to reduced 0D, 1D, 2D, and coupled hemodynamic formulations. The selected numerical model is an extended one-dimensional compliant area–flow system with variable-radius corrections, implemented in a reproducible Julia/Python benchmark package. The implementation is assessed through self-convergence, backend agreement, closure-health checks, and run-provenance records. A preliminary comparison at $t = 1.0$ s evaluates the 1D solution against two resolved 3D velocity datasets representing 23% and 40% radius reductions. Under a tetrahedral cross-section quadrature operator, the section-wise area-mean velocity discrepancies have sample-average L_1 errors of 0.505 and 0.966 cm/s, root-mean-square L_2 errors of 0.664 and 1.87 cm/s, and L_∞ maxima of 2.35 and 9.92 cm/s. The largest differences remain localized near the stronger stenosis and in reconstructed radial profiles. Because the comparison is a final-time computational cross-model diagnostic with unmatched wall, boundary, and transient histories, it is interpreted as mismatch localization rather than clinical validation or pressure-ratio evidence.

Keywords:

computational hemodynamics, blood flow simulation, stenotic vessels, Navier–Stokes, reduced-order models, idealized stenosis

Contents

1 Introduction

Coronary and arterial stenoses motivate a basic computational question: how do geometry, wall properties, rheology, and boundary data determine blood flow through a narrowed vessel? Clinical pressure-ratio measurements show why anatomical narrowing alone does not determine functional obstruction, but this report stays on the mathematical and numerical side of that problem. It studies a C^∞ idealized stenosis geometry, an extended one-dimensional area–flow realization, and a preliminary comparison with resolved 3D velocity data. The smooth geometry is a deliberate numerical device: one analytic radius profile supplies controlled derivatives, wall normals, severity parameters, and repeatable mesh families for the 1D, fixed-wall 3D Stokes, and later 3D/FSI model tiers.

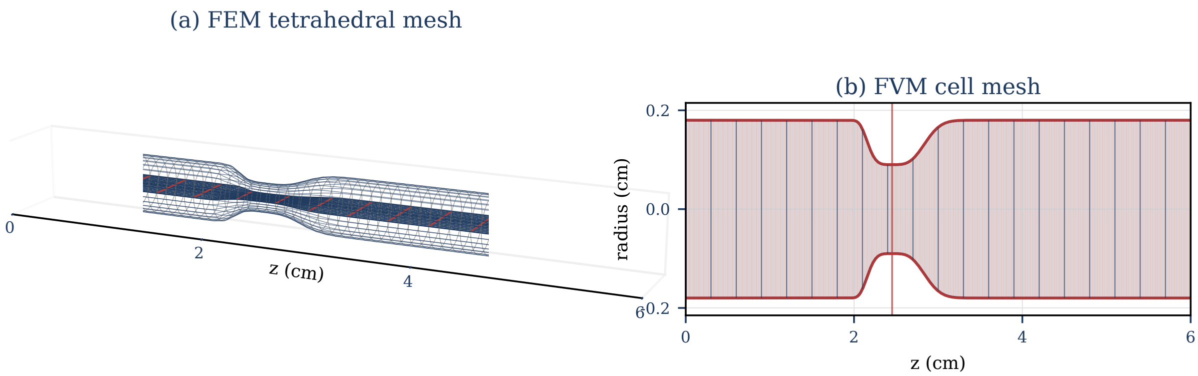

Figure 1: Finite-element and finite-volume views of the same C^∞ idealized stenosis geometry used in this report. Panel (a) shows the generated tetrahedral Gridap mesh for the stationary-Stokes initializer; panel (b) shows the one-dimensional finite-volume grid sampled along the reference radius profile $R_0(z)$. The takeaway is that one declared smooth geometry can feed distinct discretizations and controlled local refinement studies; the panels show geometry only, not velocity, pressure-drop, pressure-ratio, structural deformation, or patient-specific clinical data.

The report is organized around three answerable questions:

1. Which variables, wall law, rheology/profile closures, source terms, and boundary-state approximation define the closed stenosis-aware 1D balance law?

2. Under the recorded finite-volume/DG operators, time integrators, and package-benchmark norms, what implementation properties are verified by the reproducibility and self-convergence evidence?

3. Under the declared `CrossSectionQuadratureOperator`, what section-wise velocity and physical-flow discrepancies separate the 1D solution from the available resolved 3D velocity data?

The contribution is a consistent model-and-output specification for an idealized stenotic vessel, a reproducible implementation and benchmark suite for the selected extended 1D model, and a preliminary cross-model comparison that localizes velocity discrepancies relative to two resolved 3D computational datasets. The comparison is diagnostic rather than clinical or experimental validation.

Claim	**Evidence in this report**	**Permitted wording**
Model formulation	Derivation, closure tables, and implementation map	Defines a selected 1D model.
Smooth idealized geometry	Analytic C^∞ stenosis profile and shared finite-volume / finite-element geometry views	Supports controlled numerical studies, not patient-specific morphology.
Code behavior	Tests, self-convergence, backend checks, and manifests	Verifies reported implementation properties.
Resolved velocity comparison	Tetrahedral cross-section quadrature at $t = 1.0$ s	Reports a diagnostic cross-model comparison.
Pressure accuracy	No computed pressure comparison	Future work only.
Clinical validity	No experimental or clinical measurement comparison	No clinical validation claim.

Pressure-ratio quantities in this report are reduced model-output vocabulary, with no clinical FFR or CT-FFR measurement relationship. Convention C.3 records the admissibility requirements for any later pressure-ratio output; the completed numerical record here reports verification diagnostics and velocity-focused cross-model comparison only.

Literature Review Roadmap

- Section 1.1 fixes the continuum, kinematic, balance-law, and constitutive assumptions used throughout the report.
- Section 1.2 states the governing incompressible Navier–Stokes and generalized-Newtonian setting and then records the selected one-dimensional area–flow reduction.
- Section 1.3 places 3D, multiscale, 2D, 1D, and 0D models into a descending-dimension comparison hierarchy and clarifies what each tier can and cannot support. Section 1.4 motivates pressure-drop, pressure-ratio, and shear-output conventions from the clinical anatomy–function distinction.
- Section 2 reports implementation verification and reproducibility checks, then the quadrature-backed resolved-velocity comparison at $T = 1$. Section 3 summarizes the limitations and identifies the next evidence needed for stronger pressure–flow claims.

1.1 Continuum and Constitutive Foundations

This section moves from the macroscopic scale assumption, to continuum field variables, to blood-dynamics kinematics, to balance-law vocabulary, and finally to Newtonian versus generalized-Newtonian stress closures.

1.1.1 Physiological scale and continuum idealization

The present work is posed at the macrocirculatory arterial scale. At this scale, blood is commonly modeled as a viscous fluid whose bulk behavior can be described by continuum mechanics. The point is not to resolve individual red blood cells, platelets, or plasma constituents. Instead, the model aims to predict macroscopic observables such as flow rate, pressure, pressure drop, pressure ratio, and velocity profiles under declared assumptions. This scale separation is standard in large-vessel hemodynamic modeling, while the small-vessel and low-shear regimes require additional rheological caution [1, 2, 3].

For a final time $T_{\mathrm{final}} > 0$, let $I_T := (0, T_{\mathrm{final}})$ denote the open interior-time interval and $\overline{I}_T := [0, T_{\mathrm{final}}]$ denote the closed interval used for initial data and flow maps. Define the time-dependent blood domain and its spatial-temporal domain as:

$$\begin{cases} \Omega_B(t) := \{\boldsymbol{x} \in \mathbb{R}^3 : \boldsymbol{x} \text{ lies inside the vessel at time } t\} \\ \mathcal{Q}_T := \{(t, \boldsymbol{x}) : t \in I_T,\ \boldsymbol{x} \in \Omega_B(t)\}. \end{cases}$$

The reference and current configurations are denoted $\Omega_B(0)$ and $\Omega_B(t)$, respectively, with $\Omega_B(t)$ depending on time $t \in \overline{I}_T$. This domain notation is defined in Appendix B; boundary and function-space conventions are summarized in Appendix D. In this report, the continuum assumption is a large-vessel idealization: individual blood cells, platelets, and plasma constituents are not resolved, and blood velocity, pressure, density, and Cauchy stress are represented by sufficiently regular fields on $\mathcal{Q}_T$. Cellular-scale effects may enter only through later constitutive choices [2, Part I, Ch. 1, Secs. 1.1–1.2, pp. 4–5] [1, Secs. 1.1.3 and 3.2, pp. 12–14 and 46–47]. The definitions below fix the continuum contract used by the later model hierarchy.

Assumption 1.1 (Blood-dynamics kinematic setting)**.** For $t \in \overline{I}_T$, blood motion is represented by a sufficiently regular orientation-preserving diffeomorphism

$$\mathcal{L}_t : \Omega_B(0) \to \Omega_B(t), \quad \det \nabla_{\boldsymbol{\xi}} \mathcal{L}_t > 0$$

It carries each material label $\boldsymbol{\xi}$ to its current location $\boldsymbol{x}(t, \boldsymbol{\xi}) = \mathcal{L}_t(\boldsymbol{\xi})$. The Eulerian velocity field $\boldsymbol{u} : \mathcal{Q}_T \to \mathbb{R}^3$ satisfies

$$\partial_t \mathcal{L}_t(\boldsymbol{\xi}) = \boldsymbol{u}(t, \mathcal{L}_t(\boldsymbol{\xi})).$$

The accompanying continuum fields are pressure $p : \mathcal{Q}_T \to \mathbb{R}$, density $\rho : \mathcal{Q}_T \to \mathbb{R}^+$, and Cauchy stress $\boldsymbol{T} : \mathcal{Q}_T \to \mathbb{R}^{3\times 3}$.

Figure 2 summarizes the change of viewpoint from material labels to Eulerian field evaluation.

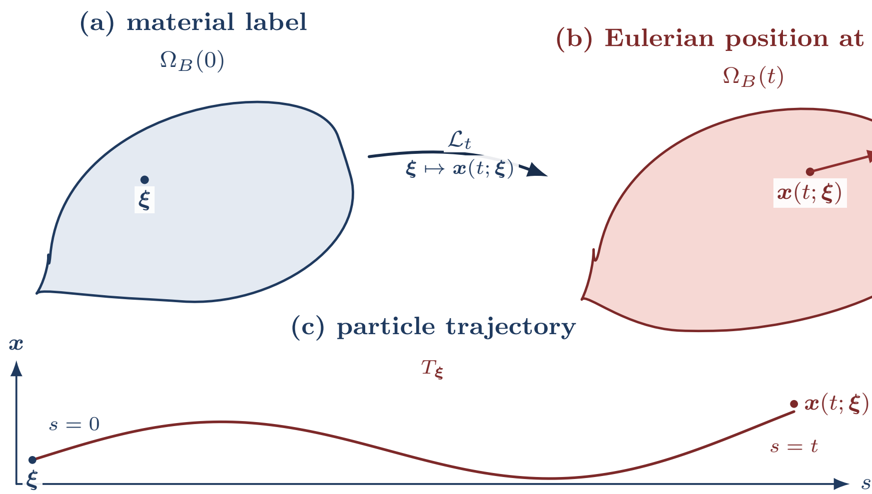

Figure 2: Lagrangian-to-Eulerian flow-map notation. The map $\mathcal{L}_t : \Omega_B(0) \to \Omega_B(t)$ sends a material label $\boldsymbol{\xi}$ to the current spatial point $\boldsymbol{x}(t;\boldsymbol{\xi}) = \mathcal{L}_t(\boldsymbol{\xi})$. The Eulerian velocity field is then evaluated at the occupied point, so the particle velocity satisfies $\partial_t \boldsymbol{x}(t;\boldsymbol{\xi}) = \boldsymbol{u}(t, \boldsymbol{x}(t;\boldsymbol{\xi})) = \boldsymbol{u}(t, \mathcal{L}_t(\boldsymbol{\xi}))$. The lower curve $T_{\boldsymbol{\xi}}$ records the same labelled particle over $s \in \overline{I}_T$, from $\boldsymbol{\xi}$ at $s = 0$ to $\boldsymbol{x}(t;\boldsymbol{\xi})$ at $s = t$.

Remark 1.2 (Eulerian vs. Lagrangian). The Eulerian description views fields such as $\phi(t, \boldsymbol{x})$ or $\boldsymbol{u}(t, \boldsymbol{x})$ at fixed spatial locations. The Lagrangian description follows material particles $\boldsymbol{\xi}$ along trajectories generated by the velocity field and views fields as functions of the material label $\boldsymbol{\xi}$ and time t, such as $\widehat{\phi}(t, \boldsymbol{\xi}) = \phi(t, \mathcal{L}_t(\boldsymbol{\xi}))$ or $\widehat{\boldsymbol{u}}(t, \boldsymbol{\xi}) = \boldsymbol{u}(t, \mathcal{L}_t(\boldsymbol{\xi}))$. The flow map $\mathcal{L}_t$ is the change of variables between these descriptions, and the trajectory of a particle with label $\boldsymbol{\xi}$ is the characteristic curve $t \mapsto \boldsymbol{x}(t;\boldsymbol{\xi}) = \mathcal{L}_t(\boldsymbol{\xi})$. Where defined, the inverse $\mathcal{L}_t^{-1}$ maps an Eulerian location back to its material label.

Classical identities in this report are read under the standing regularity convention in Convention D.1. For trajectory existence and uniqueness, the standard ODE hypothesis is that the prescribed velocity field is continuous in time and uniformly Lipschitz in space; under that condition, particle trajectories exist uniquely and depend continuously on their initial labels [4, Ch. 2].

1.1.2 Material derivative, transport, and balance laws

Definition 1.3 (Material derivative)**.** For $\phi \in C^1(\mathcal{Q}_T)$, the material derivative is the derivative along a particle trajectory:

$$D_t\phi(t, \boldsymbol{x}) \;:=\; \frac{d}{dt}\phi(t, \mathcal{L}_t(\boldsymbol{\xi})) = \partial_t\phi(t, \boldsymbol{x}) + \boldsymbol{u}(t, \boldsymbol{x}) \cdot \nabla\phi(t, \boldsymbol{x}), \qquad \boldsymbol{\xi} = \mathcal{L}_t^{-1}(\boldsymbol{x}).$$

For vector fields, the material derivative is applied componentwise. Let $\widehat{\boldsymbol{F}}_t = \nabla_{\boldsymbol{\xi}} \mathcal{L}_t$ and $\widehat{J}_t = \det \widehat{\boldsymbol{F}}_t$. In Eulerian notation this is written $\boldsymbol{F}_t(\boldsymbol{x}) = \widehat{\boldsymbol{F}}_t(\mathcal{L}_t^{-1}(\boldsymbol{x}))$ and $J_t(\boldsymbol{x}) = \det \boldsymbol{F}_t(\boldsymbol{x}) = \det \widehat{\boldsymbol{F}}_t(\mathcal{L}_t^{-1}(\boldsymbol{x}))$. The standard Jacobian evolution law is

$$D_t J_t = J_t \,\mathrm{div}\boldsymbol{u}.$$

This Jacobian identity supports the Reynolds transport theorem. Mass conservation is the separate balance law that gives the continuity equation used below; in the constant-density specialization, that equation reduces to the divergence-free constraint. Derivation details are deferred to Appendix E.

Theorem 1.4 (Reynolds Transport Theorem)**.** *Let $V(t) = \mathcal{L}_t(V(0))$ be a bounded material volume with Lipschitz boundary. If $\phi \in C^1(\mathcal{Q}_T)$, then*

$$\begin{aligned} \frac{d}{dt}\int_{V(t)} \phi \, d\boldsymbol{x} &= \int_{V(t)} (D_t\phi + \phi \, div\boldsymbol{u}) \, d\boldsymbol{x} \\ &= \int_{V(t)} (\partial_t\phi + div(\phi\boldsymbol{u})) \, d\boldsymbol{x}. \end{aligned}$$

Proof-level details for Theorem 1.4 and the momentum balance below are found in Appendix E. Assumption 1.1 supplies the 3D velocity, pressure, density, and stress fields used below. Mass conservation and linear momentum balance are imposed on material volumes and then localized to obtain the balance equations used in the model hierarchy.

Definition 1.5 (Material density preservation)**.** A flow has material density preservation if density is preserved along particle trajectories:

$$D_t\rho = 0.$$

Definition 1.6 (Incompressible flow)**.** A flow is incompressible, or volume preserving, if material volumes are preserved. With $\mathcal{L}_0 = \mathrm{Id}$ this is equivalent to

$$J_t \equiv 1 \quad \Longleftrightarrow \quad \mathrm{div}\boldsymbol{u} = 0.$$

Remark 1.7 (Divergence-free condition)*.* Mass conservation gives

$$\partial_t\rho + \mathrm{div}(\rho\boldsymbol{u}) = 0 \quad \Longleftrightarrow \quad D_t\rho + \rho\mathrm{div}\boldsymbol{u} = 0.$$

In the constant-density model used for incompressible Navier–Stokes, $\rho \equiv \rho_0 > 0$, so mass conservation reduces to $\mathrm{div}\boldsymbol{u} = 0$ in $\mathcal{Q}_T$.

Definition 1.8 (Linear momentum balance)**.** For a material fluid element $V(t) \subset \Omega_B(t)$, the linear momentum balance is

$$\frac{d}{dt}\int_{V(t)} \rho \boldsymbol{u}\, dV = \int_{\partial V(t)} \boldsymbol{T}\hat{\boldsymbol{n}}\, dS + \int_{V(t)} \rho \boldsymbol{f_b}\, dV,$$

where $\boldsymbol{T}$ is the Cauchy stress tensor and $\boldsymbol{f_b}$ is body force per unit mass.

Definition 1.9 (Strong momentum balance)**.** Under the regularity and mass-balance assumptions above, the local momentum balance is

$$\rho\left(\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) = \mathbf{div}(\boldsymbol{T}) + \rho \boldsymbol{f_b}, \qquad \text{in } \Omega_B(t).$$

The stress law closes the momentum equation. Let the rate-of-deformation tensor be

$$\boldsymbol{D}(\boldsymbol{u}) \;:=\; \frac{1}{2}\left(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^\top\right).$$

Definition 1.10. A fluid is Newtonian if, after separating the pressure contribution, its viscous stress depends linearly on the rate-of-deformation tensor $\boldsymbol{D}(\boldsymbol{u})$.

Remark 1.11 (Blood isotropy justification)*.* Blood is often modeled as an isotropic continuum fluid at large-vessel scales. This is a closure assumption for averaged behavior, not a microscopic statement about cells suspended in plasma. The cited sources support the scale and rheology caveat: nearly constant large-vessel relative viscosity versus diameter- and hematocrit-dependent apparent viscosity in small tubes, not a direct cellular-level isotropy claim [1, Sec. 1.1.3, Fig. 1.14, pp. 12–14, Sec. 3.2, pp. 46–47] [3, Abstract, pp. H1770–H1772].

Remark 1.12 (Isotropic constitutive response)*.* An isotropic fluid has a constitutive response that is independent of the chosen coordinate system. Consequently, the Cauchy stress depends on coordinate-free invariants of $\boldsymbol{D}(\boldsymbol{u})$ rather than on a particular basis.

Definition 1.13 (Newtonian, isotropic constitutive law)**.** For a Newtonian, isotropic fluid,

$$\boldsymbol{T} = -p\,\mathbf{I} + 2\eta\,\boldsymbol{D}(\boldsymbol{u}) + \lambda\,\mathrm{div}(\boldsymbol{u})\,\mathbf{I},$$

where η is the dynamic viscosity and λ is the coefficient of the volumetric-viscous term.

Definition 1.14 (Incompressible stress tensor)**.** When $\mathrm{div}\boldsymbol{u} = 0$, the Newtonian incompressible Cauchy stress is

$$\boldsymbol{T} = -p\,\mathbf{I} + 2\eta\,\boldsymbol{D}(\boldsymbol{u}).$$

Generalized-Newtonian blood models keep the stress-level constitutive form but allow the dynamic viscosity to depend on the local deformation-rate magnitude [2, Part I, Ch. 1, Secs. 2.7–3.1, pp. 16–19]. With the shear-rate convention

$$\dot{\gamma} \;:=\; \sqrt{2\boldsymbol{D}(\boldsymbol{u}) : \boldsymbol{D}(\boldsymbol{u})},$$

the incompressible generalized-Newtonian Cauchy stress is

$$\boldsymbol{T} = -p\,\mathbf{I} + 2\eta_{\text{eff}}(\dot{\gamma})\boldsymbol{D}(\boldsymbol{u}).$$

The pointwise Frobenius norm convention is

$$\|\boldsymbol{D}(\boldsymbol{u})\|^2 \;:=\; \boldsymbol{D}(\boldsymbol{u}) : \boldsymbol{D}(\boldsymbol{u}) = \sum_{i,j} D_{ij}(\boldsymbol{u})^2$$

When a source writes the viscosity as a function of $D : D$ or $|D|_{\mathrm{F}}^2$, this report translates it through the scalar shear-rate convention above. The coefficient η_{eff} is an effective dynamic viscosity; dividing by density gives the corresponding kinematic viscosity. The Newtonian incompressible model is recovered from this generalized Newtonian form when η_{eff} is constant.

Definition 1.15 (Implemented effective-viscosity closure catalog)**.** A computed 1D realization uses the rheology descriptor $\mathcal{R}$ to choose an implemented scalar effective dynamic-viscosity law $\eta_{\text{eff}}^{\mathcal{R}}(\dot{\gamma})$. The admitted descriptor values are `newtonian`, `carreau`, `carreau-yasuda`, `casson`, and `power-law`. The displayed superscripts abbreviate these descriptors, with newt, CY, and power denoting `newtonian`, `carreau-yasuda`, and `power-law`. This catalog defines the scalar dynamic-viscosity laws. A computed 1D realization evaluates the selected law on the shear-rate proxy in Appendix G and forms the closure-indexed kinematic viscosity $\nu_{\text{eff}}^{\mathcal{R}} = \eta_{\text{eff}}^{\mathcal{R}}/\rho$. Assumption 1.28 records that the velocity profile, friction law, rheology descriptor, and wall closure are part of the closed model specification. For computed non-Newtonian realizations, the laws are evaluated at the positive regularized shear rate

$$\dot{\gamma}_\epsilon \;:=\; \max\{|\dot{\gamma}|, \epsilon_\gamma\}, \qquad \epsilon_\gamma > 0, \qquad [\epsilon_\gamma] = \mathrm{s}^{-1}.$$

This positive regularization is required for the Casson and power-law families when their zero-shear limits are singular or degenerate. For descriptors that declare dynamic-viscosity bounds, the bounds satisfy $0 \le \eta_{\min} \le \eta_{\max} \le \infty$ with $\eta_{\max} > 0$ and define the interval projection

$$\Pi_{[\eta_{\min},\eta_{\max}]}(\xi) \;:=\; \min\{\eta_{\max}, \max\{\eta_{\min}, \xi\}\}.$$

The bounded viscosity is then $\Pi_{[\eta_{\min},\eta_{\max}]}(\eta_{\text{eff}}^{\mathcal{R}}(\dot{\gamma}))$; without declared bounds, this projection is the identity on the displayed model law.

$$
\begin{aligned}
\eta_{\text{eff}}^{\text{newt}} &\equiv \eta, \\
\eta_{\text{eff}}^{\text{carreau}}(\dot\gamma) &= \eta_\infty + (\eta_0 - \eta_\infty)\left(1 + (\lambda_{\text{C}}\dot\gamma_\epsilon)^2\right)^{(n_{\text{C}}-1)/2}, \\
\eta_{\text{eff}}^{\text{CY}}(\dot\gamma) &= \eta_\infty + (\eta_0 - \eta_\infty)\left(1 + (\lambda_{\text{CY}}\dot\gamma_\epsilon)^{a_{\text{Y}}}\right)^{(n_{\text{CY}}-1)/a_{\text{Y}}}, \\
\eta_{\text{eff}}^{\text{casson}}(\dot\gamma) &= \left(\sqrt{\tau_y/\dot\gamma_\epsilon} + \sqrt{\eta_p}\right)^2, \\
\eta_{\text{eff}}^{\text{power}}(\dot\gamma) &= K_{\text{pl}}\dot\gamma_\epsilon^{n_{\text{pl}}-1}.
\end{aligned}
$$

In the computed Newtonian path, the constant dynamic viscosity is $\eta = \rho\nu$, so the closure-indexed kinematic viscosity reduces to ν. The parameter domains are

$$\eta_0 > 0, \quad \eta_\infty \geq 0, \quad \eta_0 \geq \eta_\infty, \quad \lambda_{\text{C}}, \lambda_{\text{CY}} \geq 0, \quad n_{\text{C}}, n_{\text{CY}} > 0, \quad a_{\text{Y}} > 0,$$

for the Carreau and Carreau–Yasuda families,

$$\tau_y \geq 0, \qquad \eta_p > 0$$

for Casson, and

$$K_{\text{pl}} > 0, \qquad n_{\text{pl}} > 0$$

for the power-law closure. Under the SI convention, η, η_0, η_∞, η_p, $\eta_{\min}$, and $\eta_{\max}$ have units $\text{Pa}\cdot\text{s}$, λ_{C} and λ_{CY} have units s, τ_y has units Pa, and K_{pl} has units $\text{Pa}\cdot\text{s}^{n_{\text{pl}}}$. In the cgs numerical convention of Convention C.2, the corresponding dynamic-viscosity parameters are in $\text{g}/(\text{cm}\,\text{s})$, τ_y is in dyn/cm^2, and K_{pl} is in $\text{dyn}\,\text{s}^{n_{\text{pl}}}/\text{cm}^2$. The Yasuda transition exponent is denoted a_{Y} here so that it is not confused with the solver coordinate $a = R^2$ used in the reduced numerical realization. The listed families are the implemented generalized-Newtonian viscosity options used by this report's closure catalog; their parameter values are model-record data rather than universal constants [2, Part I, Ch. 1, Secs. 2.7–3.1, pp. 16–19] [5, 6].

Remark 1.16 (Newtonian blood justification). For large arteries, empirical studies suggest that apparent viscosity is often approximately constant across the relevant operating range, motivating the Newtonian approximation in many reduced-order hemodynamic models. However, diameter-dependent, hematocrit-dependent, and shear-dependent effects become increasingly important in smaller vessels and low-shear regions [1, Sec. 1.1.3, pp. 12–14] [3].

Together, Assumption 1.1 and the balance-law and stress-closure definitions in this subsection form the continuum contract used by the following hierarchy: material volumes and Reynolds transport support the mass and momentum balances, while incompressibility and the selected constitutive law specialize those balances into the PDE tiers that are later reduced by cross-sectional averaging.

1.2 Governing Equations and Selected 1D Reduction

The continuum assumptions of Section 1.1 lead to a layered modeling hierarchy. In particular, Assumption 1.1 supplies the flow map, material volumes, and continuum fields on which the conservation laws act. The high-fidelity starting point is the incompressible Navier–Stokes or generalized-Newtonian system on a three-dimensional lumen. The report then uses cross-sectional reduction to identify the selected 1D area-flow variables and to state the closure assumptions that must be fixed before any reduced numerical study is interpreted.

1.2.1 Conservation Laws and Navier–Stokes Specialization

The governing equations originate from conservation of mass and linear momentum applied to material volumes under Assumption 1.1, followed by the constant-density incompressible specialization. The derivation details and row-divergence convention are recorded in Appendix E and Definition D.6. Figure 3 summarizes the route from these balance laws to the forms used in the report [2, Part I, Ch. 1, Secs. 2.2–3.1, pp. 11–19, Part IV, Ch. 1, Sec. 1.1, pp. 380–383] [7, Sec. 6].

With this kinematic setting fixed, mass conservation for the constant-density incompressible model gives

$$\mathrm{div}\boldsymbol{u} = 0 \qquad \text{in } \Omega_B(t).$$

Linear momentum conservation balances inertia, pressure, viscous stress, and body forces:

$$\rho\left(\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) \;=\; \mathbf{div}(\boldsymbol{T}) + \rho \boldsymbol{f_b}.$$

Throughout the Navier–Stokes statements below, $\boldsymbol{f_b}$ denotes body force per unit mass, so $\rho\boldsymbol{f_b}$ is the corresponding body force per unit volume.

Definition 1.17 (Conservative incompressible Navier–Stokes)**.** On $\Omega_B(t)$, the constant-density Newtonian system may be written in conservative tensor-divergence form as

$$\begin{cases} \partial_t(\rho\boldsymbol{u}) + \mathbf{div}(\rho\boldsymbol{u}\otimes\boldsymbol{u}) = -\nabla p + \mathbf{div}(2\eta\boldsymbol{D}(\boldsymbol{u})) + \rho\boldsymbol{f_b}, \\ \mathrm{div}\boldsymbol{u} = 0, \qquad \rho \equiv \rho_0 > 0. \end{cases}$$

For constant density and divergence-free velocity, $\mathbf{div}(\rho\boldsymbol{u}\otimes\boldsymbol{u}) = \rho(\boldsymbol{u}\cdot\nabla)\boldsymbol{u}$. Thus the same Newtonian momentum equation can be read in advective form.

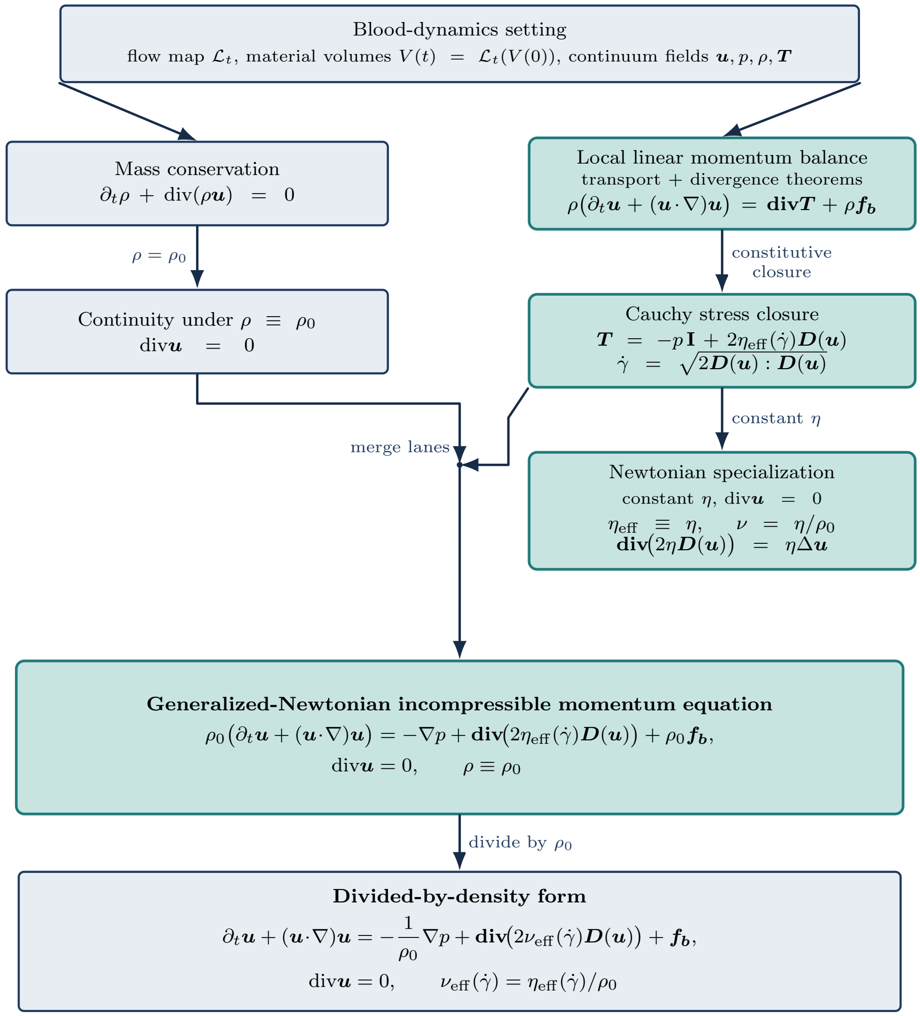

Figure 3: From the blood-dynamics setting to incompressible Navier–Stokes. The flow-map and material-volume assumptions support the conservation statements. The mass-conservation/incompressibility lane and the stress-closure lane merge into the generalized-Newtonian incompressible momentum form. The Newtonian box is a constant-viscosity specialization, and the divided-by-density form records the corresponding kinematic-viscosity scaling.

Definition 1.18 (Generalized-Newtonian incompressible Navier–Stokes)**.** For an incompressible generalized-Newtonian fluid, use the shear-rate convention

$$\dot\gamma := \sqrt{2\boldsymbol{D}(\boldsymbol{u}) : \boldsymbol{D}(\boldsymbol{u})}.$$

The velocity $\boldsymbol{u}$, pressure p, body force per unit mass $\boldsymbol{f_b}$, and constant density $\rho \equiv \rho_0 > 0$ satisfy

$$\begin{cases} \rho\big(\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\big) = -\nabla p + \mathbf{div}\Big(2\eta_{\text{eff}}(\dot\gamma)\boldsymbol{D}(\boldsymbol{u})\Big) + \rho \boldsymbol{f_b}, \\ \operatorname{div}\boldsymbol{u} = 0. \end{cases}$$

Definition 1.19 (Density-scaled generalized Navier–Stokes)**.** With $\nu_{\text{eff}}(\dot\gamma) := \eta_{\text{eff}}(\dot\gamma)/\rho$, the divided-by-density form is

$$\begin{cases} \partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u} = -\rho^{-1}\nabla p + \mathbf{div}\Big(2\nu_{\mathrm{eff}}(\dot\gamma)\boldsymbol{D}(\boldsymbol{u})\Big) + \boldsymbol{f_b}, \\ \mathrm{div}\boldsymbol{u} = 0. \end{cases}$$

The density-scaled form is particularly useful for numerical simulations, because it separates the convective ($(\boldsymbol{u}\cdot\nabla)\boldsymbol{u}$) and diffusive / viscous-stress ($\mathbf{div}(2\nu_{\mathrm{eff}}(\dot\gamma)\boldsymbol{D}(\boldsymbol{u}))$) contributions in the momentum equation, allowing for more straightforward numerical treatment of each term independently.

Definition 1.20 (Newtonian incompressible Navier–Stokes)**.** If $\eta_{\mathrm{eff}} \equiv \eta$ is constant, then sufficiently smooth divergence-free velocity fields satisfy $\mathbf{div}(2\eta\boldsymbol{D}(\boldsymbol{u})) = \eta\Delta\boldsymbol{u}$. The Newtonian equations are

$$\begin{cases} \rho\big(\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\big) = -\nabla p + \eta\Delta\boldsymbol{u} + \rho\boldsymbol{f_b}, \\ \mathrm{div}\boldsymbol{u} = 0, \end{cases} \qquad \begin{cases} \partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u} = -\rho^{-1}\nabla p + \nu\Delta\boldsymbol{u} + \boldsymbol{f_b}, \\ \mathrm{div}\boldsymbol{u} = 0, \end{cases}$$

where $\nu \ := \ \eta/\rho$.

Thus the 3D PDE tier is not introduced independently: flow maps and material volumes support Reynolds transport, Reynolds transport gives mass and momentum balance, the Cauchy stress law closes momentum, and incompressibility selects the Navier–Stokes systems above.

Remark 1.21 (Navier–Stokes system classification)*.* Navier–Stokes is a nonlinear coupled PDE system. The unknowns are the three velocity components and pressure. The convective term transports momentum, the pressure gradient enforces incompressibility, and the viscous term diffuses momentum through the stress tensor.

1.2.2 Boundary Data and Well-Posedness Context

For a single-vessel three-dimensional domain, write the boundary decomposition

$$\partial\Omega_B(t) = \Gamma_{\mathrm{in}}(t) \cup \Gamma_{\mathrm{out}}(t) \cup \Gamma_w(t).$$

Here $\Gamma_{\mathrm{in}}(t)$ denotes an inlet portion where a velocity, flow-rate, or profile condition may be prescribed. The outlet portion $\Gamma_{\mathrm{out}}(t)$ carries pressure, traction, or reduced downstream network data. For a fixed-wall no-slip continuum model, the wall portion $\Gamma_w(t) \subset \partial\Omega_B(t)$ carries $\boldsymbol{u} = 0$; this condition is not imposed on the inlet or outlet portions of $\partial\Omega_B(t)$. Other continuum or FSI tiers may use no-penetration, wall-law, or structure-coupling data instead. A complete initial-boundary value problem also states $\boldsymbol{u}(0,\cdot) = \boldsymbol{u}_0$, the domain, forcing, compatibility assumptions, and the solution class.

Remark 1.22 (Global regularity context)*.* For smooth, spatially localized initial data in three dimensions, global existence and smoothness for Navier–Stokes remains an open Clay Millennium Prize problem [8, Problem statement]. This is mathematical context for the 3D continuum model, not a direct limitation of the selected reduced 1D model introduced below.

Well-posedness results for a three-dimensional Navier–Stokes initial-boundary value problem are problem-specific: the statement must fix the domain class, boundary conditions, forcing, compatibility assumptions, initial-data space, solution class, and time horizon. This report uses that vocabulary only as PDE background for the model hierarchy. It does not use the Clay global-regularity problem as support for a local theorem, and it does not prove a well-posedness result for the moving arterial domain or for the reduced numerical solver. The domain and function-space notation relevant to such statements is summarized in Appendix D; coordinate and energy background is collected in Appendix F.

1.2.3 Cross-Sectional Reduction and Selected 1D Model

The 1D model reduces the spatial dimension by assuming that the vessel has a distinguished axial direction and that pressure and velocity can be averaged over cross-sections. This keeps pulse-wave and pressure-drop structure while discarding secondary flow, separation, recirculation, and resolved wall traction. The reduction follows standard compliant-artery and dimension-reduced modeling arguments [9, Sec. 2, pp. 253–257] [1, Ch. 3, Secs. 3.2 and 3.6] [10, Sec. 3, pp. 24–26]. The reduced variables A, Q, $\bar{u}$, and p_{g} should therefore be read as cross-sectional outputs of the continuum fields under additional averaging, geometry, wall-law, and profile assumptions, not as independent replacements for the blood-dynamics setting above.

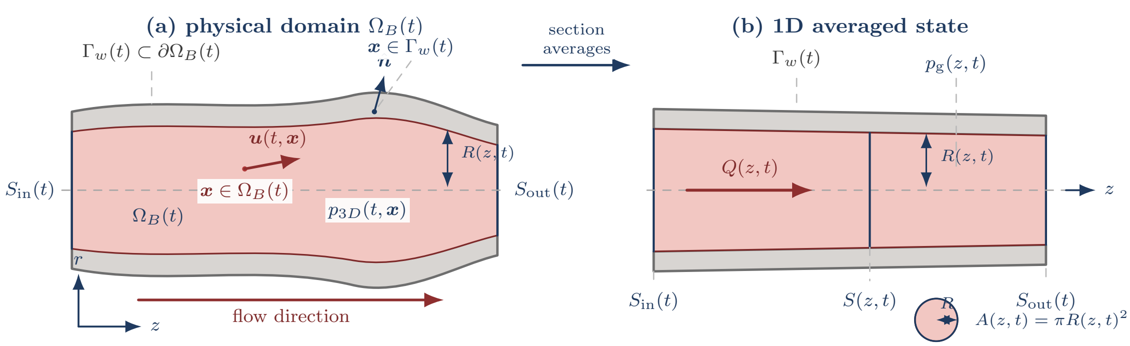

Figure 4: Compliant-vessel notation for the 1D reduction. The current configuration $\Omega_B(t)$ is represented by a straightened averaging domain: section averages over $S(z,t)$ define $A(z,t)$, $Q(z,t)$, $\bar{u}(z,t)$, and $p_{\mathrm{g}}(z,t)$, while $R(z,t)$ records the radius entering the wall and geometry closures. The figure is a bookkeeping reduction from continuum fields to retained 1D variables, not a resolved wall-traction or secondary-flow model.

Assumption 1.23 (Straight single-vessel geometry)**.** The selected 1D model assumes a vessel of length L aligned with the z-axis and a circular cross-section with radius $R(z,t)$. For $0 < z < L$, $S(z,t)$ is the interior cross-section used for averaging, not a boundary component of the open lumen.

Definition 1.24 (Section-averaged 1D fields)**.** The retained reduced variables are

$$\begin{aligned}
A(z,t) &:= \int_{S(z,t)} 1\, dS, && \text{cross-sectional area,}\\
Q(z,t) &:= \int_{S(z,t)} u_z\, dS, && \text{volumetric flow rate,}\\
\bar{u}(z,t) &:= \frac{Q(z,t)}{A(z,t)}, && \text{mean axial velocity,}\\
p_{\mathrm{g}}(z,t) &:= \frac{1}{A(z,t)} \int_{S(z,t)} p_{3D,\mathrm{g}}\, dS, && \text{mean gauge pressure.}
\end{aligned}$$

Here $p_{3D,\mathrm{g}}$ is continuum pressure in the wall-law gauge frame. The pressure-reference convention for reduced ratios is fixed separately in Convention C.3.

Definition 1.25 (Wall and coupling closure axes)**.** The wall descriptor $\mathcal{W}$ records how the wall or wall-reduced mechanics are closed. The coupling descriptor $\mathcal{C}$ records whether the single vessel is isolated or coupled to another model tier. Both are independent of the rheology descriptor $\mathcal{R}$: $\mathcal{R}$ changes the stress or effective-viscosity law, $\mathcal{W}$ changes the wall mechanics, and $\mathcal{C}$ changes the interface or network contract.

Axis	**Descriptor**	**Information retained**	**Reading in this report**
Wall	$\mathcal{W}_{\mathrm{rigid}}$	Fixed or prescribed wall geometry; no structural state is solved	A fixed-wall continuum or rigid-tube reduced baseline.
Wall	$\mathcal{W}_{\mathrm{elastic1D}}$	Dynamic area $A(z,t)$, gauge pressure $p_{\mathrm{g}}(A,z,t)$, reference area $A_0(z)$, wall stiffness $\beta(z)$, and external pressure	A pressure–area wall-law family. The selected implementation uses its Canic–Koiter member; no separate structural PDE is solved.
Wall	$\mathcal{W}_{\mathrm{resolvedFSI}}$	Fluid velocity and pressure plus wall displacement, wall velocity, structural stress, and interface conditions	A full moving-wall FSI PDE system, used here only as comparison vocabulary.
Coupling	$\mathcal{C}_{\mathrm{isolated}}$	Inlet and outlet data close one vessel	The baseline numerical study.
Coupling	$\mathcal{C}_{\mathrm{network}}$ or $\mathcal{C}_{\mathrm{3D/1D}}$	Interface pressure, flow, traction, impedance, or characteristic data connecting 0D, 1D, and 3D tiers	A separate boundary/interface contract, not a wall law.

This separation follows the standard distinction between reduced compliant vessel models, geometrical multiscale coupling, and resolved FSI models [9, Sec. 2, pp. 253–257] [11] [12].

Definition 1.26 (Classical no-slip 1D baseline)**.** The classical no-slip 1D baseline used as a model-family member starts from the fixed-wall continuum trace $\boldsymbol{u} = 0$ on the wall boundary $\Gamma_w \subset \partial\Omega_B$, together with separate inlet and outlet data on Γ_{in} and Γ_{out}. After cross-sectional averaging, the no-slip antecedent is represented by the parabolic profile/friction convention $\alpha_{\mathcal{P}} = 4/3$ and $g_{\mathcal{P}} = 4$. In the implementation manifests this member is recorded as `classical-1d-no-slip`; it keeps the same reduced state variables and selected Canic–Koiter pressure–area wall law as the extended 1D framework, but omits the Canic effective-alpha variable-radius correction. The descriptor `canic-extended-1d` records the current extended-model member.

Definition 1.27 (Closed 1D area-flow model)**.** For $\mathcal{W} = \mathcal{W}_{\text{elastic1D}}$ and an incompressible fluid with density ρ, the working compliant single-vessel model is written with the composite axial pressure derivative

$$\frac{d}{dz}p_{\text{g}}(A(z,t),z,t) = \partial_A p_{\text{g}}(A,z,t)\partial_z A + \partial_z p_{\text{g}}(A,z,t)|_A \,.$$

The reduced balance law is

$$\partial_t A + \partial_z Q = 0, \tag{1.1}$$

$$\partial_t Q + \partial_z \left(\alpha \frac{Q^2}{A}\right) + \frac{A}{\rho}\frac{d}{dz}p_{\text{g}}(A(z,t),z,t) = -C_f(z,A,Q)\frac{Q}{A}. \tag{1.2}$$

The parameter α is the momentum-flux correction factor induced by the selected cross-sectional velocity profile, C_f is the wall-friction closure, and $p_{\text{g}}(A,z,t)$ is the wall-law gauge pressure closing the reduced system. This is the reduced $\mathcal{W}_{\text{elastic1D}}$ member of the wall-model axis: wall response is represented through the pressure–area law rather than through a separate structural PDE [9, Sec. 2, pp. 253–255] [1, Ch. 3, Sec. 3.2, pp. 40–47].

Assumption 1.28 (Velocity-profile and rheology closure)**.** A closed 1D area-flow model must declare the momentum-flux correction factor α, the wall-friction closure C_f, any rheology descriptor $\mathcal{R}$, and the pressure–area wall closure $\mathcal{W}$. These choices determine how cross-sectional profile effects, wall friction, viscosity or effective viscosity, and wall compliance enter the reduced equations. Numerical-record details such as solver coordinates, shear-rate proxies, and discrete source terms are deferred to Appendix G.

Definition 1.29 (Selected Canic–Koiter pressure–area wall law)**.** The selected elastic wall closure is the Canic–Koiter thin-wall pressure–area law. In conventional area notation it may be written

$$p_{\text{g}}(A,z,t) = p_{\text{ext}}(z,t) + \frac{\beta(z)}{A_0(z)}\left(\sqrt{A(z,t)} - \sqrt{A_0(z)}\right). \tag{1.3}$$

Here $A_0(z)$ is the reference area, p_{ext} is the external pressure, and $\beta(z)$ is a wall-stiffness parameter. Some implementations absorb A_0^{-1} into β; the essential closure is that p_{g} is known once A_0, β, and p_{ext} are supplied. In the radius-squared coordinate $a = R^2$ used by the implemented solver, the selected member is

$$p_{1,\text{g}}(a,z,t) = p_{\text{ext}}(z,t) + \frac{K}{R_0(z)^2}\left(\sqrt{a} - R_0(z)\right), \qquad K = \frac{Eh}{1-\sigma^2}.$$

The diagnostic extended pressure then adds the Canic variable-radius correction

$$p_{2,\mathrm{g}}(a, Q, z, t) = g_{\mathcal{P}} \rho \nu_{\mathrm{eff}}^{\mathcal{R}} \frac{Q}{a} \frac{R_0'(z)}{R_0(z)}.$$

This p_2 term is a variable-radius viscous correction, not a replacement wall law. Resolved FSI would replace the algebraic pressure–area closure with structural unknowns and fluid–structure interface conditions.

Definition 1.30 (Physical-to-solver map for the implemented 1D state)**.** The continuum 1D model uses physical area A_{phys} and physical flow Q_{phys}. The implemented radius-squared solver instead stores

$$a = R^2, \qquad q = \frac{Q_{\mathrm{phys}}}{\pi}.$$

Thus

$$A_{\mathrm{phys}} = \pi a, \qquad Q_{\mathrm{phys}} = \pi q, \qquad \bar{u} = \frac{Q_{\mathrm{phys}}}{A_{\mathrm{phys}}} = \frac{q}{a}.$$

With $K = Eh/(1 - \sigma^2)$ and the Canic conservative-form convention $R_0^* = R_{\max}$, the selected wall law gives the solver elastic flux contribution recorded in Appendix G.3:

$$\Psi_h(a) = \frac{K}{3\rho R_{\max}^2} a^{3/2}, \qquad \partial_a \Psi_h(a) = \frac{K}{2\rho R_{\max}^2} \sqrt{a}.$$

This map is the convention used by the Section 2 velocity and flow comparisons.

Definition 1.31 (Stenosis reference geometry)**.** A stenosis is encoded through the reference radius or reference area. Let $R_{\mathrm{base}}(z)$ be the nominal healthy radius and $s(z) \in [0, 1)$ a fractional radius-reduction profile. Then

$$R_0(z) = R_{\mathrm{base}}(z)\big(1 - s(z)\big), \qquad A_0(z) = \pi R_0(z)^2. \tag{1.4}$$

The baseline idealized stenotic-vessel profile is the smooth asymmetric kernel

$$g(z) = z - z_s + \kappa \exp\left(-\frac{(z - z_a)^2}{2\sigma_a^2}\right), \qquad \psi(z) = \exp\left(-\lambda g(z)^4\right), \tag{1.5}$$

$$s(z) = s_{\max} \psi(z), \qquad 0 \le s_{\max} < 1. \tag{1.6}$$

In the default centimeter-scale simulation record, $z_a = 2.5\,\mathrm{cm}$, $z_s = 3.4\,\mathrm{cm}$, $\sigma_a = 1\,\mathrm{cm}$, $\kappa = 0.95\,\mathrm{cm}$, and $\lambda = 50\,\mathrm{cm}^{-4}$. This geometry is the baseline idealized simulation geometry in this report, not a patient-specific plaque morphology or clinical calibration.

Remark 1.32 (Stenosis-profile and variable-radius boundary). The profile in Eq. 1.6 is C^∞ and rapidly localized rather than compactly supported. This smooth idealized geometry is the baseline for the reduced

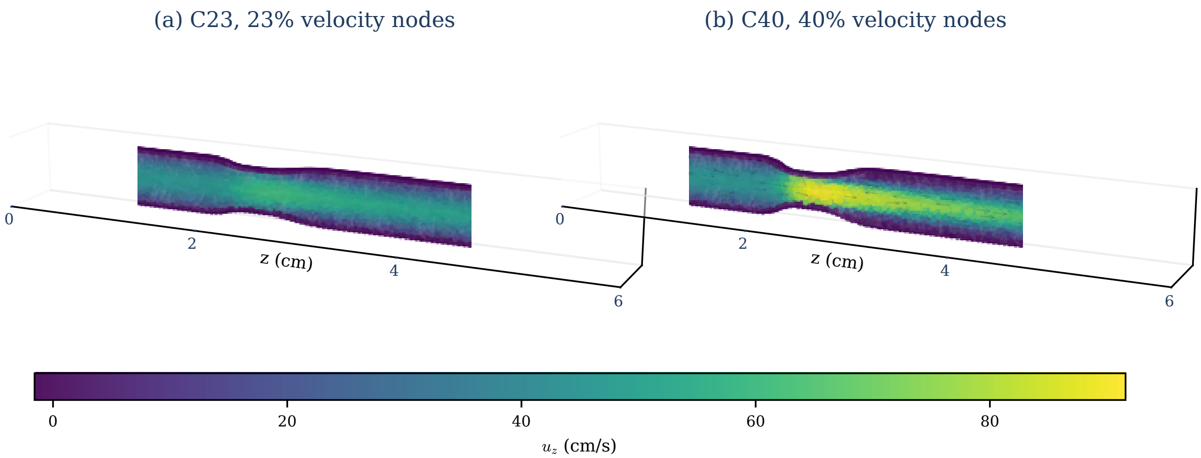

Figure 5: Resolved 3D node-centered axial velocity fields for the two $T = 1$ comparison cases used in Section 2. Panel (a) shows C23 with 22,691 loaded nodes; panel (b) shows C40 with 21,975 loaded nodes. The shared color scale makes the stronger axial-velocity concentration in C40 visible. Sparse arrows indicate local vector direction, and the transparent surface frames the 3D point cloud. These are resolved velocity-data renderings, not the stationary-Stokes pathline proxy or a pressure-drop, FFR, or patient-specific clinical calibration measurement.

simulations in this report and is also useful for fixed-wall 3D studies: the same analytic radius profile supplies controlled derivatives, wall normals, severity variation, and repeatable mesh-refinement families without introducing segmentation noise or patient-specific morphology. Canic et al. show that standard 1D variable-geometry models may miss stenotic pressure and velocity trends unless additional variable-radius terms are included [13, Abstract, Secs. 1–2]. Those corrections are included only by the `canic-extended-1d` forward-model descriptor; `classical-1d-no-slip` retains the same selected wall law but omits the Canic variable-radius corrections.

Definition 1.33 (Conservative flux/source notation for the reduced wall law)**.** For a reduced wall closure $\mathcal{W}$ admitting an elastic potential, define

$$\mathbf{U} = (A, Q)^\top, \qquad \mathbf{F}_{\mathcal{W}}(\mathbf{U}, z, t) = \begin{pmatrix} Q \\ \alpha Q^2/A + \Psi_{\mathcal{W}}(A, z, t) \end{pmatrix},$$

$$\mathbf{S}_{\mathcal{W},\mathcal{R}}(\mathbf{U}, z, t) = \begin{pmatrix} 0 \\ S^{\mathcal{W}}_{\mathrm{geom}}(A, z, t) + S^{\mathcal{R}}_{f}(A, Q, z) \end{pmatrix}.$$

The potential and sources are tied to the selected closure descriptors by

$$\begin{aligned}
\partial_A \Psi_{\mathcal{W}}(A,z,t) &= \frac{A}{\rho}\partial_A p_{\mathrm{g}}^{\mathcal{W}}(A,z,t),\\
S_{\mathrm{geom}}^{\mathcal{W}}(A,z,t) &= \partial_z \Psi_{\mathcal{W}}(A,z,t)\big|_A - \frac{A}{\rho}\,\partial_z p_{\mathrm{g}}^{\mathcal{W}}(A,z,t)\big|_A\,,\\
S_f^{\mathcal{R}}(A,Q,z) &= -C_f^{\mathcal{R}}(z,A,Q)\frac{Q}{A}.
\end{aligned}$$

For $\mathcal{W}_{\mathrm{elastic1D}}$, $p_{\mathrm{g}}^{\mathcal{W}}$ is the wall law in Definition 1.29. In component form,

$$\partial_t A + \partial_z Q = 0, \tag{1.7}$$

$$\partial_t Q + \partial_z\left(\alpha\frac{Q^2}{A} + \Psi_{\mathcal{W}}(A,z,t)\right) = S_{\mathrm{geom}}^{\mathcal{W}}(A,z,t) + S_f^{\mathcal{R}}(A,Q,z), \tag{1.8}$$

The wall descriptor $\mathcal{W}$ determines the pressure–area law, elastic potential, wave speed, and geometry source; the rheology descriptor $\mathcal{R}$ determines the viscosity or friction closure. A resolved FSI model is therefore not obtained by merely relabeling S_{geom}: it introduces structural state and interface laws. The displayed split records the balance-law objects needed by finite-volume schemes, while a computed realization may still use a particular well-balanced or source-split discretization [14] [15, Summary, Secs. 2.1–2.3, Secs. 5–6] [12].

Definition 1.34 (Single-vessel boundary data)**.** For a single-vessel solve on $z \in [0, L]$, prescribe one inlet condition and one outlet condition:

$$\begin{aligned}
&Q(0,t) = Q_{\mathrm{in}}(t) \quad \text{or} \quad \bar{u}(0,t) = \bar{u}_{\mathrm{in}}(t),\\
&p_{\mathrm{g}}(L,t) = p_{\mathrm{out,g}}(t) \quad \text{or} \quad p_{\mathrm{g}}(L,t) = p_{\mathrm{WK,g}}(Q(L,\cdot),t).
\end{aligned}$$

The second outlet option denotes a terminal Windkessel-style relation. The numerical study uses prescribed $Q_{\mathrm{in}}(t)$ and $p_{\mathrm{out,g}}(t)$ because they give an unambiguous reduced single-vessel contract.

Definition 1.35 (Characteristic speeds for the reduced elastic subsystem)**.** Let $\lambda_\pm$ denote the two characteristic speeds of the positive-area elastic 1D subsystem obtained from the homogeneous (A, Q) balance law. Their formula and assumptions are given in Appendix G.2.

Assumption 1.36 (Subcritical boundary regime)**.** The baseline boundary treatment is used only in the subcritical regime $\lambda_- < 0 < \lambda_+$, where one characteristic enters through the inlet and one characteristic enters through the outlet.

Under this regime, a finite-volume scheme constrains the incoming information at each boundary and leaves the outgoing state determined by the interior solve. This is the reduced-model analogue of declaring enough boundary data for the selected equation class; it is not, by itself, a validation of a particular boundary discretization.

1.3 Model Hierarchy and Comparison Boundary

Reduced hemodynamic models form a hierarchy of modeling roles rather than a scalar accuracy ranking. The governing equations above start from a three-dimensional continuum problem and then introduce reduced descriptions, so this subsection reads the hierarchy in descending spatial dimension. Dimension, wall treatment, rheology, boundary data, coupling laws, output operators, and numerical realization remain separate modeling choices. This report selects the 1D area–flow model as the reduced forward map; 3D, multiscale, 2D, and 0D models supply comparison, coupling, diagnostic, and boundary-condition vocabulary.

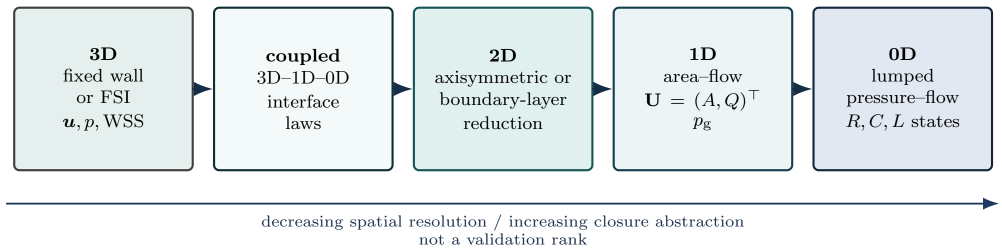

Figure 6: Model hierarchy for reading comparison claims, ordered from the three-dimensional continuum tier toward reduced and lumped closures. The horizontal direction is a dimensional-reduction and closure-abstraction axis, not a validation ranking. This report selects a 1D reduced forward model; 3D fixed-wall, resolved FSI, clinical, and independent-solver comparisons require separately matched assumptions, inputs, output operators, and discrepancy metrics before they can support evidence claims.

1.3.1 3D Fixed-Wall and Resolved-FSI Models

The three-dimensional tier is the natural starting point after the governing equations because it keeps the lumen domain, velocity field $\boldsymbol{u}(t, \boldsymbol{x})$, pressure field $p(t, \boldsymbol{x})$, stress tensor, and boundary decomposition of the continuum problem. In a fixed-wall hemodynamic model, the wall geometry is fixed or prescribed while the fluid equations are solved on the resolved lumen [7]. This tier can supply resolved velocity, pressure, and wall traction. A WSS output additionally requires the stress tensor, wall normal, wall location, tangent convention, and output operator to be declared [16]. These quantities are fields on the resolved lumen and describe the vessel wall's interaction with the fluid domain, not the 1D state variables A, Q, and p_g.

For the idealized stenosis used here, a fixed-wall stationary-Stokes study is the locally feasible 3D tier. The C^∞ radius profile gives an analytic wall and repeatable tetrahedral mesh families, so resolved velocity, pressure, and stress-derived wall-traction diagnostics can be studied without changing the wall shape in time. This is still a fixed-wall fluid solve: it does not recover wall displacement, wall velocity, structural stress, or interface work.

Resolved FSI changes the wall descriptor rather than merely adding a new output. A fixed-wall model belongs to the $\mathcal{W}_{\mathrm{rigid}}$ or prescribed-wall reading in Definition 1.25; a resolved-FSI model uses $\mathcal{W}_{\mathrm{resolvedFSI}}$ and adds wall displacement, wall velocity, structural stress, interface kinematics, and fluid–structure force balance. The elastic 1D wall law $\mathcal{W}_{\text{elastic1D}}$ is a reduced pressure–area closure, not a structural FSI solve [2, 12].

The 3D/FSI tier can represent eccentricity, curvature, branches, secondary flow, separation, and recirculation more directly than a 1D area–flow model. That role does not make it assumption-free. A valid comparison requires matched geometry, rheology descriptor $\mathcal{R}$, wall closure $\mathcal{W}$, boundary data, mesh and time resolution, output operators, and discrepancy metrics. A velocity-slab diagnostic, a traction-based WSS comparison, and a pressure-ratio comparison are different observation problems.

The 3D/FSI tier supplies the reference vocabulary for resolved velocity, pressure, traction, and wall motion, but it does not automatically validate a reduced model without matched comparison data. It is not the selected broad parametric generator in this report because the setup and computational burden are high, not because resolved 3D or FSI models are unimportant.

1.3.2 Multiscale Coupling

A multiscale model is a coupled forward problem that combines tiers of the hierarchy. A concrete 3D–1D–0D example is a resolved local 3D stenosis segment embedded between upstream and downstream 1D arterial vessels, with terminal 0D RCR or Windkessel beds closing the distal circulation. The purpose is to keep local resolved physics where it matters while representing pulse-wave propagation and distal impedance without making the entire circulation a 3D domain [11].

The interface laws are part of the model. At a 3D–1D interface, the record must state which 3D cross-section is coupled, how 3D velocity and traction are reduced to 1D flow or mean pressure, and whether compatibility is imposed through pressure, normal traction, total flow, characteristic variables, or a stabilized coupling variant. At a 1D–0D interface, terminal 1D flow enters the ODE load and the 0D circuit returns a pressure, impedance, or characteristic boundary relation. Resolved-FSI coupling additionally requires compatible wall kinematics, structural stresses, and interface work. These choices affect stability and interpretation, and they must be recorded with the numerical coupling convention [12].

A multiscale model is not a new fidelity tier; it is a coupled forward problem whose interface laws are part of the model definition. Coupling more components does not automatically make a comparison more correct. The boundary data, interface variables, time coupling, output maps, and discrepancy metrics decide what claim the computation can support. In this report, multiscale coupling provides comparison and boundary-condition vocabulary; it is not the baseline single-vessel solve unless explicitly declared.

1.3.3 2D Intermediate Resolved-Flow Models

A 2D model is treated here as a conceptual intermediate resolved-flow tier, usually posed as an axisymmetric or planar flow problem. It retains a spatially resolved velocity and pressure field on an idealized reduced domain rather than only the section-averaged state (A, Q, p_{g}). Within those idealizations, it can inspect radial velocity structure, throat acceleration, simplified recirculation, and wall-shear trends that a 1D area–flow model cannot resolve.

The same idealization also fixes the boundary of the evidence. Axisymmetric and planar models suppress eccentric plaques, curvature, torsion, branch geometry, non-axisymmetric secondary flow, and fully patient-specific WSS. Their wall outputs depend on the declared stress tensor, wall location, normal, tangent, and output operator, just as in the shear convention of Convention C.4. A 2D result is therefore not interchangeable with matched 3D or resolved-FSI evidence. The broad continuum, pipe-flow, and dimension-reduction background for this reading is supplied by [1, 2].

The 2D tier is a diagnostic bridge: useful for controlled mechanism checks, but too idealized to serve as clinical or resolved-3D evidence. In this report it can help interpret whether a reduced trend is plausibly tied to throat acceleration, radial profile structure, or simplified recirculation, but it is not the selected forward generator and does not validate the 1D map by itself.

1.3.4 1D Distributed Area–Flow Models

The selected 1D reduction is defined in Section 1.2.3. There the retained fields $A(z,t)$, $Q(z,t)$, $\bar{u}(z,t) = Q/A$, and $p_{\mathrm{g}}(z,t)$, the wall and coupling closure catalog, the elastic-wall area–flow balance law, the stenosis reference geometry, the conservative flux/source notation, and the single-vessel boundary data are recorded in Definitions 1.24–1.34 [1, 9]. This subsection uses that machinery only to identify the hierarchy role of the 1D tier.

In the hierarchy, the key point is that the 1D state is distributed but reduced. It retains axial area, flow, mean velocity, and gauge pressure; it does not retain the full three-dimensional velocity field, pressure field, secondary flow, separation, recirculation, or wall traction. Reduced shear and viscosity quantities therefore inherit the declared velocity-profile, wall-friction, and rheology closures. Tube-flow and hemorheology data caution that diameter, hematocrit, and shear-rate dependence cannot be promoted to resolved local physics without a declared closure map [2, 3]. Reduced shear proxies are not traction-based WSS unless a separate observation map declares how a reduced output is compared with a resolved wall-stress quantity.

The closure contract remains part of the model definition. A 1D case must declare the momentum-flux/profile convention, friction law, rheology descriptor $\mathcal{R}$, wall descriptor $\mathcal{W}$, coupling descriptor $\mathcal{C}$, boundary data, numerical realization, and output operators. Stenosis enters through the reference geometry $R_0(z)$, $A_0(z)$, and $s(z)$, but resolved throat flow does not appear automatically; variable-radius corrections such as the Canic-style terms are additional closure choices [13]. The current implementation records this axis explicitly through the `canic-extended-1d` and `classical-1d-no-slip` forward-model descriptors. Both use the selected `canic-koiter-thin-membrane` pressure–area wall law; the former adds the Canic variable-radius corrections, while the latter corresponds to the wall no-slip baseline in Definition 1.26.

The 1D model is the selected reduced forward map because it retains distributed pressure–flow structure at low computational cost, while its outputs remain reduced quantities with declared observation maps. In this report it is the selected single-vessel generator for the numerical study, not a clinical-validation claim, an independent-solver parity claim, or a resolved-3D-accuracy claim.

1.3.5 0D Outlet and Network Models

A 0D model is a lumped ODE or circuit model for pressures, flows, volumes, and loads. It retains compartment pressures, branch flows, resistances, compliances, and, when the circuit includes inertial effects, inertances. The model preserves a time-dependent pressure–flow relation for parts of the circulation that are not spatially resolved. Windkessel and RCR models are the canonical terminal examples: they supply an outlet impedance, downstream load, or network closure for a local 1D, multiscale, or 3D vessel solve [1, 10].

A 0D model closes the circulation seen by the local vessel model; it does not resolve the local stenosis. The spatial coordinate has been integrated out, so the model has no local stenosis geometry, no axial pressure-drop field, no velocity profile, no throat acceleration, and no recirculation region. It also has no resolved wall-shear stress or wall traction, because it does not carry a lumen surface, a stress tensor on that surface, or a wall-normal output operator.

The 0D tier is therefore useful boundary and network context for this report. It can prescribe an outlet load, organize distal-bed assumptions, or provide a lumped circulation closure. It is not the selected local stenosis-resolution tier and cannot replace the distributed area–flow state used in the numerical study in this report.

A comparison claim requires matched assumptions and declared output operators: the continuum setting, geometry, rheology descriptor $\mathcal{R}$, wall descriptor $\mathcal{W}$, coupling descriptor $\mathcal{C}$, boundary data, numerical settings, observation maps, and discrepancy metrics must all be part of the comparison record. This report treats the selected 1D area–flow model as the forward map for the numerical study. Resolved-3D/FSI agreement, clinical calibration, and independent-solver parity are later evidence standards, not assumptions built into the model hierarchy. The pressure–flow motivation is stated after the continuum and model-hierarchy setup so that pressure-drop, pressure-ratio, and shear quantities are introduced as model outputs with declared conventions, not as clinical measurements or validation targets.

1.4 Clinical Pressure–Flow Motivation

Anatomy–function distinction. Coronary stenosis is the motivating lesion class because a visible narrowing does not automatically imply a flow-limiting obstruction. Clinical assessment therefore distinguishes anatomical stenosis detection from functional pressure–flow assessment [17, p. 4401] [18, Chs. 1–2 and 5] [19, Sec. 1]. Coronary lesion-imaging studies can describe local geometry and lesion length, but those anatomical descriptors do not by themselves define a pressure–flow output [20, Abstract, Methods]. This anatomy–function split motivates reporting pressure drop and reduced pressure ratio under declared observation and reference conventions, rather than treating stenosis severity $s_{\max}$ alone as the reported reduced output. Section 2 reports velocity diagnostics only; the pressure-drop and pressure-ratio conventions fix reduced-output vocabulary for the model framework. Figure 7 summarizes this motivation as a schematic distinction between visible narrowing and pressure–flow loss.

Clinical fractional flow reserve (FFR) is an invasive pressure-wire index, conventionally interpreted in its measurement setting as distal coronary pressure divided by aortic pressure during maximal hyperemia [18, Ch. 5, Sec. 5.5] [21, Abstract]. FAME lesion-level evidence documented the discordance between angiographic and functional severity and is the clinical source for the familiar FFR ≤ 0.80 decision context [22, Abstract]. Computed tomography-derived FFR (CT-FFR) studies supply noninvasive comparison vocabulary by inferring pressure–flow quantities from coronary computed tomography angiography (CCTA)-derived anatomy and comparing them against invasive FFR or related reference measurements [23, Abstract, Methods] [19, Secs. 1.1–1.2]. Those clinical settings motivate the output vocabulary but are not reproduced by the completed numerical study in this report. The reduced pressure ratio PR_{1D} is therefore a convention-governed model output, not a reported FFR, CT-FFR, or pressure-measurement comparison.

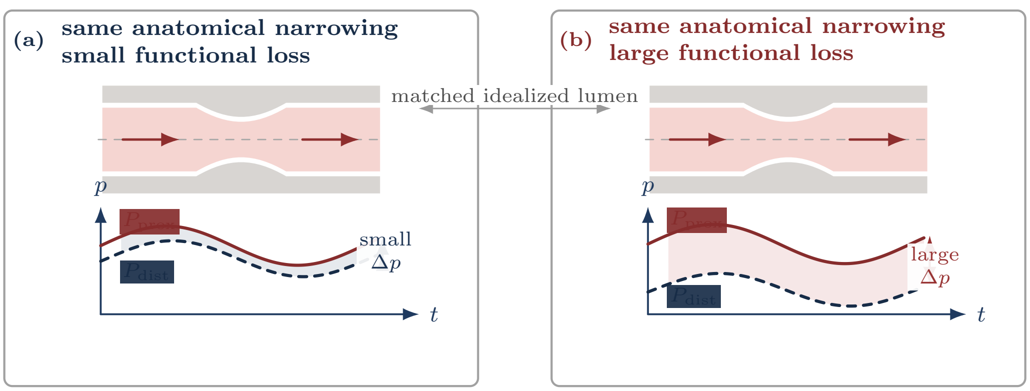

Figure 7: Pressure–flow motivation for reduced pressure outputs. The same anatomical narrowing can correspond to either a small functional pressure loss in panel (a) or a larger loss in panel (b). The reduced-output target is therefore the declared pressure-loss signature, not stenosis severity alone.

For the selected model gauge-pressure field p_{g}, define the observed proximal and distal pressure traces through declared scalar observation maps:

$$P_{\mathrm{prox}}(t) \;:=\; \mathcal{O}_{\mathrm{prox}}[p_{\mathrm{g}}(\cdot, t)], \qquad P_{\mathrm{dist}}(t) \;:=\; \mathcal{O}_{\mathrm{dist}}[p_{\mathrm{g}}(\cdot, t)].$$

The pressure-drop trace is invariant under common additive pressure shifts:

$$\Delta p(t) \;:=\; P_{\mathrm{prox}}(t) - P_{\mathrm{dist}}(t).$$

Unlike $\Delta p(t)$, a pressure ratio depends on the pressure reference used before forming the numerator and denominator. A reduced pressure-ratio output must therefore declare its proximal and distal observation maps, pressure reference, averaging window, and denominator admissibility. Convention C.3 records those requirements and the trace-level definition of $\mathrm{PR}_{1D}(\pi_{\mathrm{PR}})$. Its high-flow averaging window is a recorded output window, not a clinical hyperemia model unless an external measurement comparison is separately specified. Figure 8 shows the corresponding axial-tap shorthand for an idealized 1D stenosis.

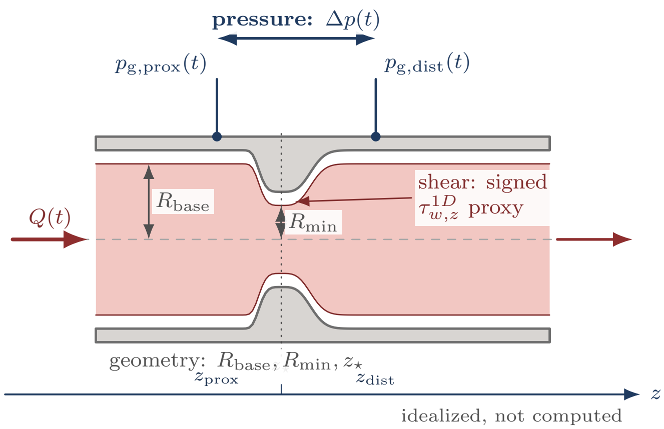

Figure 8: Schematic tap and output convention for an idealized stenosis, not drawn from a computed case. The blue pressure taps define $\Delta p(t)$, the gray geometry landmarks locate the stenosis region, and the red/orange cue marks the reduced shear-proxy convention. Convention C.3 supplies the pressure-ratio observation maps, averaging window, pressure reference, and admissibility rule for $\mathrm{PR}_{1D}(\pi_{\mathrm{PR}})$.

Shear diagnostics. Pressure drops and pressure ratios are lesion-scale pressure-output conventions. A resolved wall-shear-stress output belongs to a resolved stress state and must declare the stress tensor, wall normal, wall location, output operator, and signed tangent direction when a signed component is reported [16, Abstract, Sec. 2, pp. 8–10]. Reduced shear proxies are model-tier diagnostics rather than clinical anatomy–function outputs; Convention C.4 records the declaration requirements for comparisons with traction-based wall shear stress.

2 Verification, Reproducibility, and Cross-Model Velocity Comparison

This section first records the reproducibility checks for the solver package and then reports the final-time resolved-velocity comparison. The comparison is velocity-focused: pressure-drop, pressure-ratio, FFR, and clinical validation claims remain outside the completed numerical evidence.

2.1 Verification and Reproducibility

The report package includes a reproducible package-benchmark pipeline for the Julia solver package and Python CLI. The suite records descriptor health, self-convergence, native/SciML agreement, rheology/profile sensitivity, OpenBF-style boundary handling, resolved-velocity diagnostics, and Python CPU/MPS checks for the emitted CSV records. These are internal implementation and reproducibility checks. They do not become clinical validation, pressure or FFR accuracy evidence, full transient FEM validation, or scientific equivalence beyond the metrics actually compared.

```
./scripts/julia-release simulations/run_package_benchmark.jl \
  --profile overnight \
  --output-dir simulations/output/package_benchmark/<run_id> \
  --overwrite \
  --include-python \
  --include-resolved3d \
  --publish-report-assets

pipenv run python scripts/render_package_benchmark_figures.py \
  --benchmark-dir simulations/output/package_benchmark/<run_id>
```

The smoke profile is a deterministic wiring check. The overnight profile expands the same schemas to the descriptor-health, refinement, native/SciML parity, stationary-Stokes projection, rheology/profile, OpenBF-style boundary, resolved-velocity, and Python CPU/Torch-MPS comparison stages. Optional SciPy and resolved-3D inputs are treated as availability boundaries: absent inputs produce structured skipped rows rather than manuscript claims.

Table 1: Package benchmark stage summary.

Stage	Rows	OK	Skipped/error
case results	18	18	0
refinement	128	128	0
backend parity	18	18	0
stokes ic	12	12	0
rheology profile	60	60	0
boundary openbf	3	3	0
resolved3d	4	4	0
python mps	9	9	0

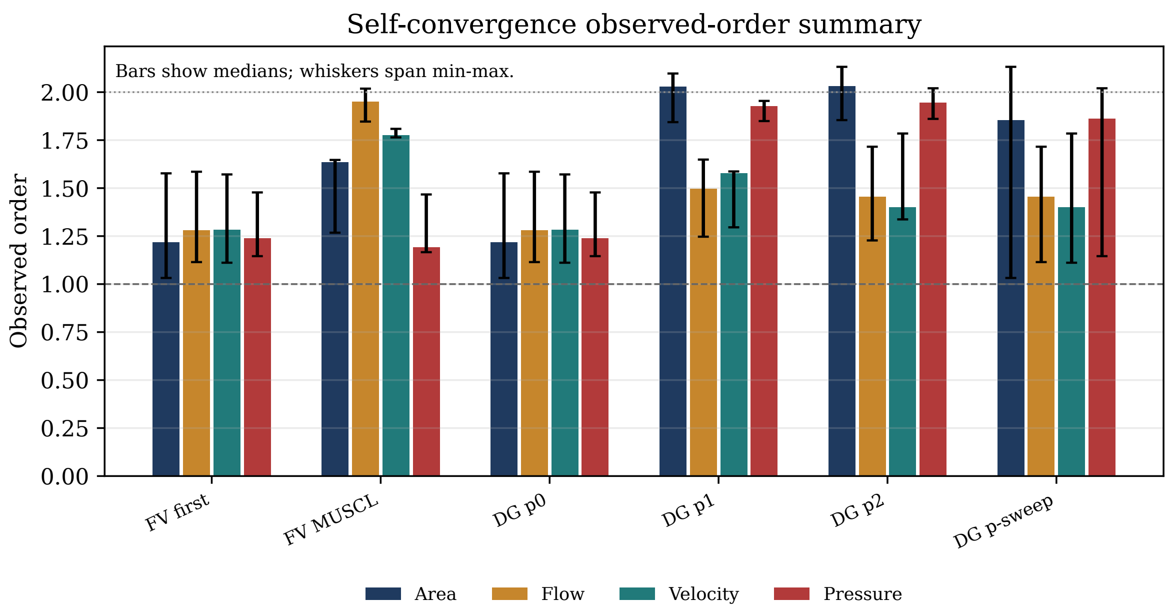

Figure 9: Self-convergence observed-order summary emitted by the package benchmark renderer. Bars report median observed order by method and metric; whiskers show the observed min–max range across the benchmarked refinements. This is an internal convergence diagnostic, not external validation evidence.

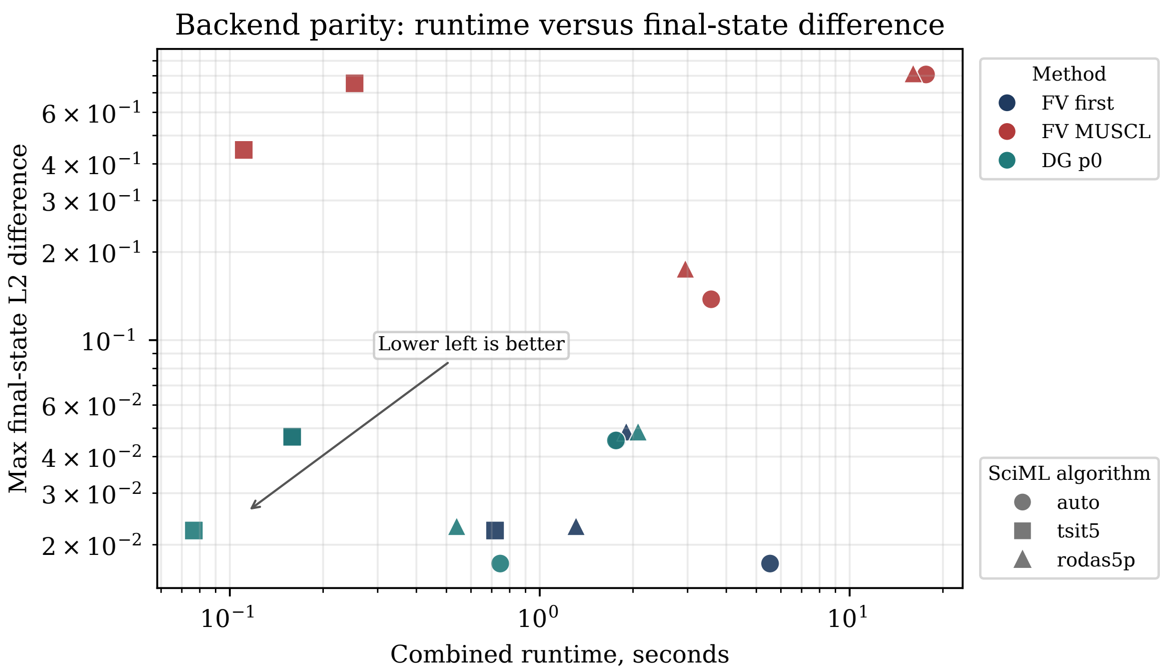

Figure 10: Native/SciML backend agreement versus runtime for benchmarked compatible methods. Color identifies the method family, marker shape identifies the SciML algorithm, and lower-left points are both faster and closer to the native final state under the plotted aggregate difference.

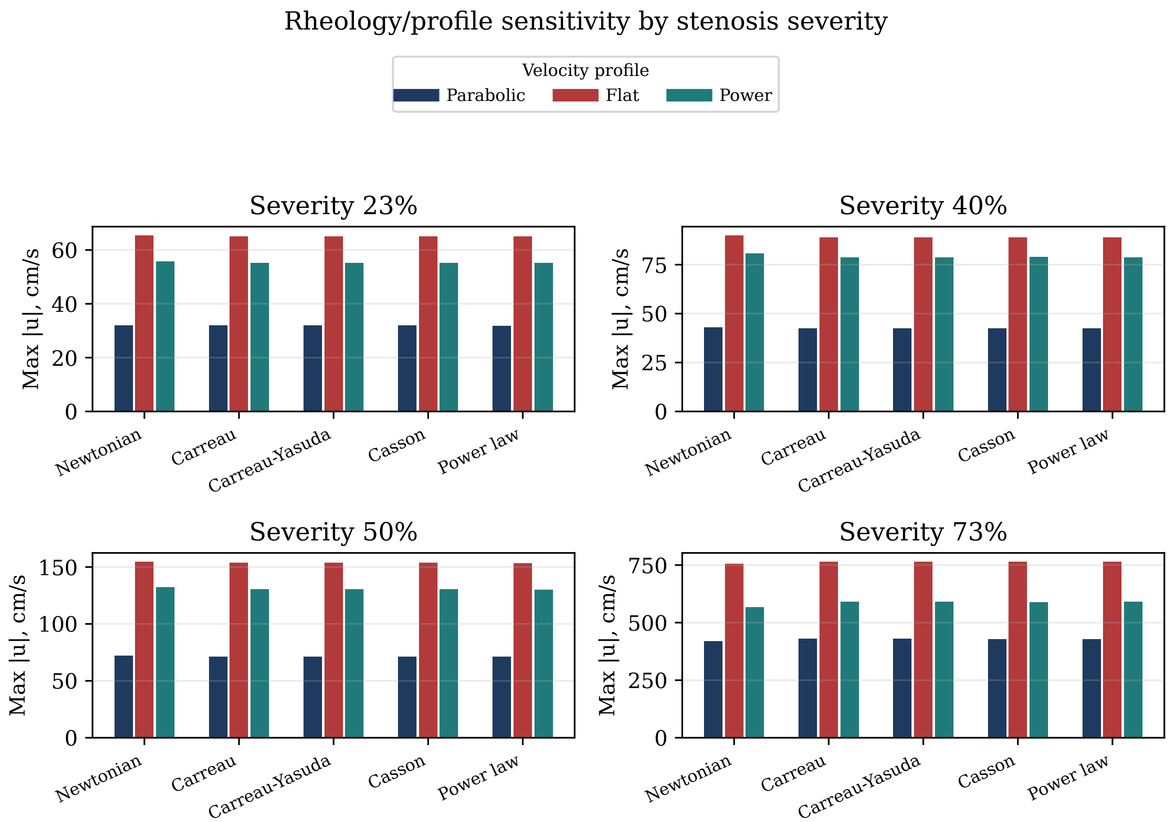

Figure 11: Rheology/profile sensitivity summary for declared descriptor combinations. Facets separate stenosis severity; grouped bars separate the rheology and velocity-profile descriptors. The diagnostic shows how the maximum speed responds to descriptor choices within the emitted benchmark records.

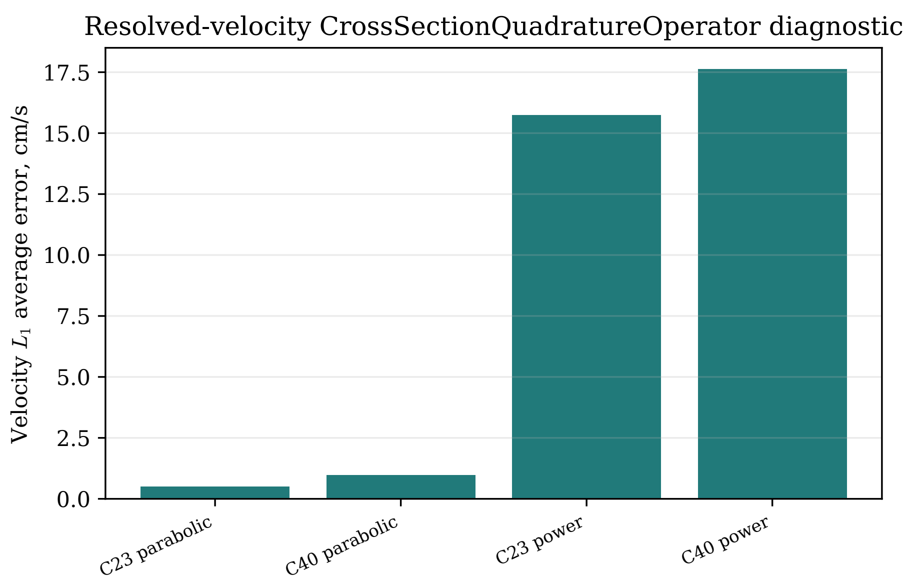

Figure 12: Resolved-velocity output-operator diagnostic summary for the available local cases. The rendered value is the section-wise velocity L_1 average error for the declared `CrossSectionQuadratureOperator`.

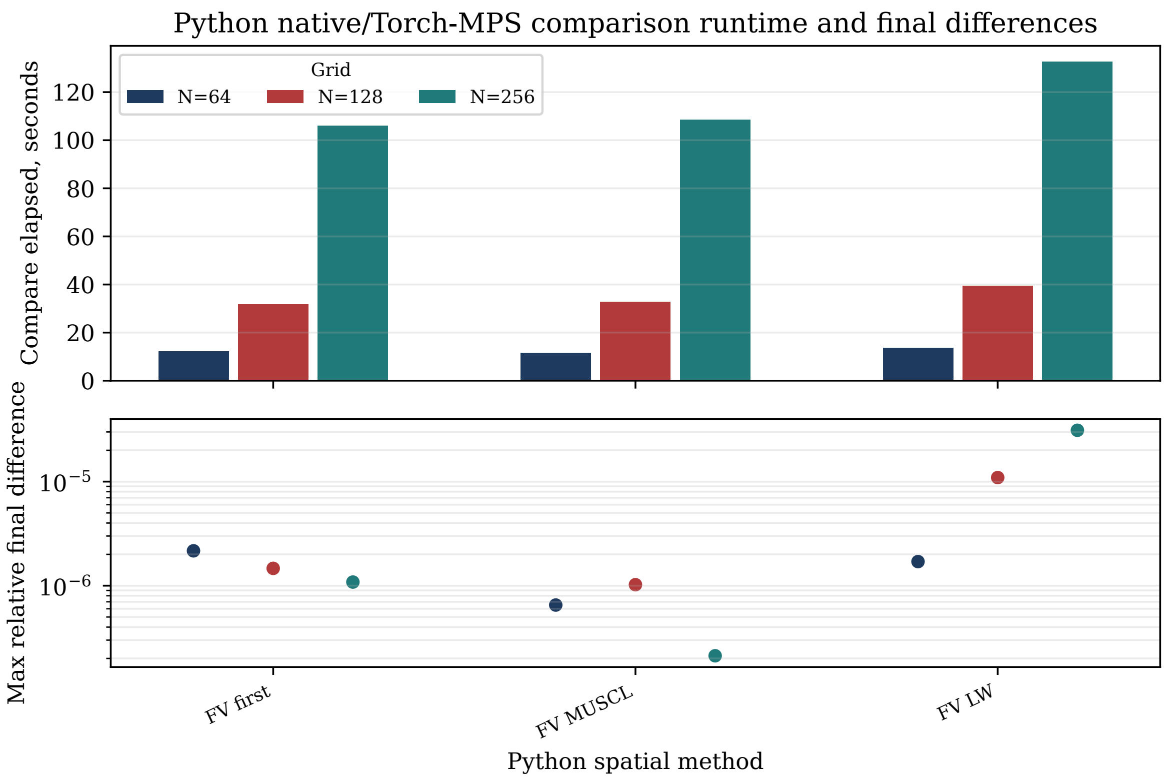

Figure 13: Python native CPU versus Torch-MPS comparison for the Python CLI surface. The top panel reports elapsed time by method and grid size; the lower panel reports the maximum relative final-state difference across the emitted area and flow summaries.

2.2 Preliminary Cross-Model Velocity Comparison

The resolved-velocity comparison evaluates the selected 1D solution against two available resolved 3D velocity datasets at $t = 1.0$ s. The cases are labelled C23 and C40 by stenosis severity. They are computational cross-model diagnostics, not pressure-flow results and not clinical validation.

Solver-coordinate convention. In this section, the stored solver state `A` is $a = R^2$, not physical circular area. The physical map is Definition 1.30: $A_{\text{phys}} = \pi a$, $Q_{\text{phys}} = \pi q$, and $\bar{u} = q/a = Q_{\text{phys}}/A_{\text{phys}}$.

All comparison numbers come from `simulations/output/3d_comparison/full_t1_native_nx400/`, interpreted under Convention G.1. The regeneration command, source-path convention, and file hashes are listed in Appendix H. The raw XDMF/HDF5 inputs remain local ignored files under `simulations/data/3d/canic_case3/`; the report archives the generated CSV and PGFPlots-ready static data.

For a target plane $S_j = \Omega_{3D} \cap \{z = z_j\}$, the default operator listed in Table 2 intersects every tetrahedron with S_j, linearly interpolates axial velocity on the cut edges, sorts the cut polygon vertices in the plane, and triangulates the polygon by a centroid fan. Let $\mathcal{Q}_j$ denote the resulting cut triangles. The reported 3D area,

Table 2: Case matching for the resolved-velocity comparison. The case labels C23 and C40 are used in the main text and figures; the source ID records the upstream local data directory.

Label	**Source ID**	**Velocity metadata**	**Operator**	**3D time (s)**	**Time error (s)**
C23	77	`77/velocity.xdmf`	`CrossSectionQuadratureOperator`	0.9994999999999453	5.0×10^{-4}
C40	60	`60/velocity.xdmf`	`CrossSectionQuadratureOperator`	0.9994999999999453	5.0×10^{-4}

Table 3: Shared 1D realization and output-operator parameters for the $t = 1.0$ comparison. Per-step diagnostics such as realized maximum CFL, positivity events, and residual summaries are not persisted in the available comparison artifact.

Quantity	**Recorded value**
Forward model	`canic-extended-1d`.
Wall model	$\mathcal{W}_{\text{elastic1D}}$ with the selected Canic–Koiter pressure–area law `canic-koiter-thin-membrane`.
Rheology	Newtonian rheology with $\nu = 0.04$ cm^2/s.
Velocity profile	Parabolic profile with $\alpha_{\mathcal{P}} = 4/3$.
Variable-radius terms	Canic effective-alpha and geometry-source corrections.
Discretization	$N = 400$, $L = 6$ cm, MUSCL finite volume with minmod limiter, native SSPRK3 time integration.
Time controls	$T = 1.0$ s, timestep cap 10^{-5} s, CFL cap 0.45.
Initial and boundary state	Geometry-rest initial state $a = R_0^2$, $q = 0$; inlet $\bar{u}_{\text{in}} = 22.5$ cm/s; outlet $a_{\text{out}} = R_0(L)^2$ with the implemented characteristic approximation in Appendix G.5.
Section targets	200 equally spaced axial planes on $0 \le z \le 6$ cm.
Radial targets	Slices $z = 1.951$, 2.451, and 2.951 cm with 20 bins in $r/R_0(z)$.
Supplemental sensitivity	Node-slab arithmetic means are emitted only to `node-slab-sensitivity.csv` for half-widths 0.0075, 0.0150753769, and 0.0301507538 cm.

flow, and mean velocity are

$$A_{3D,j} = \sum_{\Delta \in \mathcal{Q}_j} |\Delta|, \qquad Q_{3D,j} = \sum_{\Delta \in \mathcal{Q}_j} |\Delta| \, \bar{u}_{z,\Delta}, \qquad \bar{u}_{3D,j} = Q_{3D,j}/A_{3D,j}.$$

For a cut triangle Δ whose vertices carry reconstructed axial velocities $u_{z,1}$, $u_{z,2}$, and $u_{z,3}$,

$$\bar{u}_{z,\Delta} = \frac{u_{z,1} + u_{z,2} + u_{z,3}}{3}.$$

The corresponding 1D quantities use physical flow, not the raw solver flow coordinate:

$$Q_{1D,j} = \pi q(z_j, t), \qquad \bar{u}_{1D,j} = q(z_j, t)/a(z_j, t).$$

Velocity and flow differences are reported with L_1, L_2, and L_∞ summaries over valid section targets. With $e_{u,j} = \bar{u}_{1D,j} - \bar{u}_{3D,j}$, the velocity norms are

$$\|e_u\|_{1,\text{avg}} = \frac{1}{n} \sum_j |e_{u,j}|, \qquad \|e_u\|_{2,\text{avg}} = \left(\frac{1}{n} \sum_j |e_{u,j}|^2 \right)^{1/2}, \qquad \|e_u\|_\infty = \max_j |e_{u,j}|,$$

$$\operatorname{rel} L_1 = \frac{\sum_j |e_{u,j}|}{\sum_j |\bar{u}_{3D,j}|}, \qquad \operatorname{rel} L_2 = \frac{\left(\sum_j |e_{u,j}|^2\right)^{1/2}}{\left(\sum_j |\bar{u}_{3D,j}|^2\right)^{1/2}}.$$

Radial profile rows use the same cut triangles, binned by triangle-centroid radius. Empty bins are reported as invalid rows rather than zero-velocity observations.

The same quadrature record audits whether the cut operator is sampling the expected static cross-section. With the reference area $A_{\mathrm{ref}}(z_j) = \pi R_0(z_j)^2$, define

$$\epsilon_A(z_j) = \frac{|A_{3D}(z_j) - A_{\mathrm{ref}}(z_j)|}{A_{\mathrm{ref}}(z_j)}.$$

Table 4 reports this geometry/operator audit for the same 200 section targets. It verifies that the tetrahedral cut areas agree with the expected fixed stenosis cross-sections to sub-percent typical discrepancy, with the largest differences confined to a few section targets.

Table 4: Quadrature cut-area audit against the static reference $A_{\mathrm{ref}}(z) = \pi R_0(z)^2$. Discrepancies are percentages over valid section targets and are emitted in the tracked `area-audit.dat` report asset.

Case	**Sections**	**min ϵ_A**	**median ϵ_A**	**mean ϵ_A**	**max ϵ_A**
C23	200	0.045	0.281	0.389	2.94
C40	200	0.005	0.291	0.397	4.46

Table 5: Quadrature-backed section velocity and physical-flow summary at $t = 1.0$ s. Mean velocities and velocity errors are in cm/s; physical flows and flow errors are in cm^3/s. The 1D physical flow is $Q_{1D} = \pi q$.

Case	**Sections**	$\langle \bar{u}_{1D} \rangle$	$\langle \bar{u}_{3D} \rangle$	$\langle Q_{1D} \rangle$	$\langle Q_{3D} \rangle$	$L_1(u)$	$L_2(u)$	**rel.** L_1	**rel.** L_2	$L_1(Q)$	$L_2(Q)$
C23	200	24.11	23.63	2.288	2.241	0.505	0.664	0.0214	0.0277	0.0482	0.0509
C40	200	26.43	25.52	2.287	2.219	0.966	1.87	0.0379	0.0688	0.0709	0.0893

Table 6: Generated solver diagnostics for the same runs. The positive characteristic-radicand values are in solver units of cm^2/s^2 and support the reported characteristic-speed evaluation on the sampled final state.

Case	**min α_{eff}**	**max α_{eff}**	**min χ^2** (cm^2/s^2)	**Valid sections**
C23	1.33058	1.33333	8.15×10^5	200
C40	1.32500	1.33333	6.36×10^5	200

Figure 14 shows the area-mean axial velocity comparison. The 1D and quadrature-averaged 3D means agree closely away from the largest acceleration zone, while the largest mismatch remains localized near the stronger stenosis. The C40 maximum section error is 9.92 cm/s at $z = 2.261$ cm; C23 reaches 2.35 cm/s at $z = 2.291$ cm.

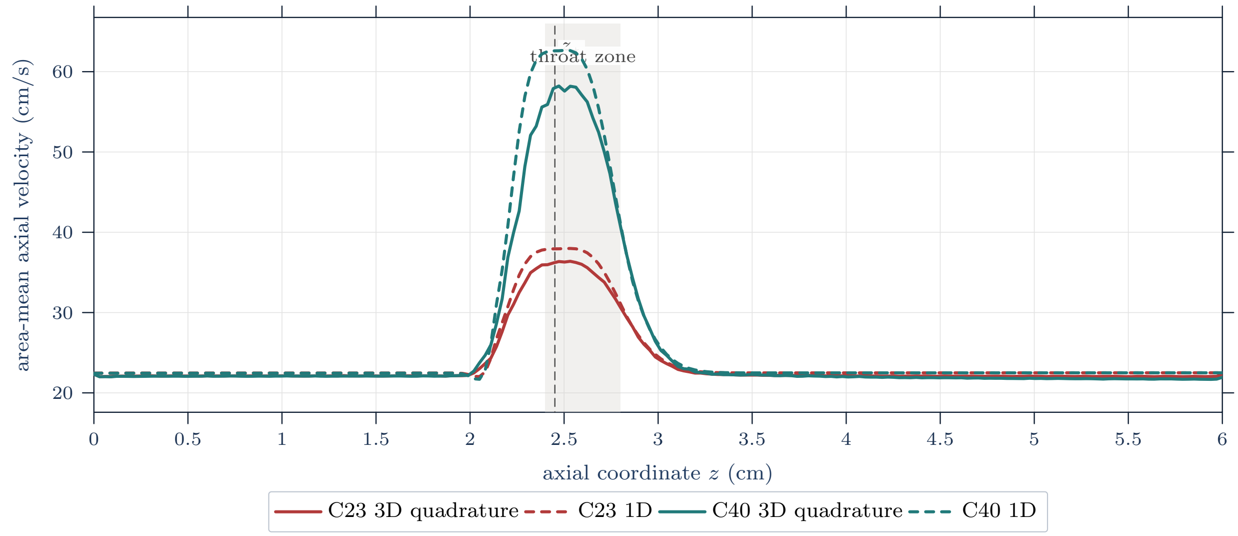

Figure 14: Area-mean axial velocity comparison at $t = 1.0$ s. Solid curves are tetrahedral cross-section quadrature means from the resolved 3D velocity field; dashed curves are interpolated 1D means $\bar{u} = q/a = Q_{\text{phys}}/A_{\text{phys}}$. The shaded band marks the throat-centered stenosis region.

Table 7: Five largest section-wise velocity errors across the two included cases. Velocities and velocity errors are in cm/s; physical flows and flow errors are in cm^3/s, with $Q_{1D} = \pi q$.

Rank	**Case**	z **(cm)**	$\bar{u}_{1D}$	$\bar{u}_{3D}$	$\lvert e_u \rvert$	Q_{1D}	Q_{3D}	$\lvert e_Q \rvert$	**Tri.**
1	C40	2.261	52.58	42.66	9.92	2.201	1.862	0.339	759
2	C40	2.291	56.94	48.20	8.74	2.230	1.903	0.327	749
3	C40	2.352	61.43	53.22	8.21	2.270	1.980	0.290	743
4	C40	2.322	59.81	52.09	7.72	2.253	1.970	0.283	771
5	C40	2.231	46.95	39.94	7.01	2.174	1.918	0.256	833

The radial-profile reconstruction in Figure 15 is a secondary diagnostic. It asks whether the 1D velocity-profile closure reconstructed from the section mean resembles area-weighted radial bins of the resolved 3D cut. The throat and downstream slices show the largest C40 discrepancy, which is consistent with the section-mean localization and with the fact that the 1D profile closure cannot represent secondary flow, skewness, or separation.

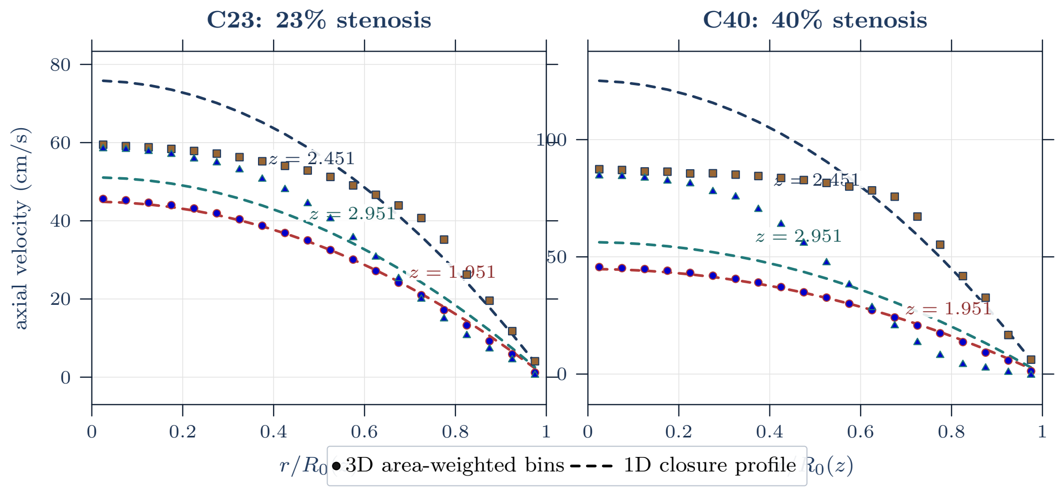

Figure 15: Radial velocity profile comparison at the declared throat-centered slices. Markers are resolved 3D area-weighted radial-bin means from the quadrature triangles; dashed curves are the 1D closure profile reconstructed from q/a. Colored in-panel labels identify the axial stations.

Table 8: Radial-profile discrepancy by case and slice. Bin means are area-weighted over the quadrature triangles in each bin; the displayed L_1 and L_2 summaries are unweighted over the valid radial-bin means. Velocity errors are in cm/s; radial-bin physical-flow errors are in cm^3/s and use $Q_{1D,\mathrm{bin}} = \bar{u}_{1D,\mathrm{bin}} A_{\mathrm{bin}}$. All listed slices have 20 valid radial bins.

Case	z (cm)	$L_1(u_r)$	$L_2(u_r)$	$L_\infty(u_r)$	$L_1(Q_r)$	$L_2(Q_r)$
C23	1.951	0.493	0.614	1.34	0.00290	0.00498
C23	2.451	7.47	9.47	16.35	0.0130	0.0152
C23	2.951	5.15	5.62	7.62	0.0191	0.0229
C40	1.951	0.561	0.649	1.41	0.00303	0.00478
C40	2.451	16.79	21.78	37.73	0.0163	0.0220
C40	2.951	16.47	18.89	28.36	0.0495	0.0568

These discrepancies should be read in light of the comparison boundary in Section 1.3. The quadrature operator gives a physical cross-sectional average of the resolved velocity field at the matched final time, but the comparison still changes model tier, wall treatment, boundary realization, and transient history. The result therefore localizes velocity mismatch; it does not attribute that mismatch uniquely to the 1D closure and does not supply pressure, FFR, or clinical accuracy evidence. The next defensible pressure-focused study would need declared pressure observations and a matched evidence record, not only this velocity diagnostic.

3 Conclusions and Limitations

This report specified a velocity-focused mathematical and numerical study for an idealized stenotic vessel. The continuum material-flow setting, balance laws, stress closure, Navier–Stokes specialization, and 1D area–flow reduction define the modeling vocabulary. The selected implementation is a Canic-extended, radius-squared 1D compliant-vessel solver with declared wall, rheology, velocity-profile, boundary, and output conventions.

The implementation evidence answers the reproducibility question, not a clinical one. The package benchmark suite records descriptor health, self-convergence, native/SciML agreement, stationary-Stokes initialization checks, boundary diagnostics, rheology/profile sensitivity, resolved-velocity diagnostics, and Python CPU/MPS comparisons for the emitted artifacts. The C^{∞} stenosis profile is useful here because it supplies one analytic geometry for the 1D solver, the fixed-wall stationary-Stokes mesh, and future 3D/FSI mesh-refinement studies. Those records support the reported implementation properties and benchmark figures; they do not establish pressure accuracy, full transient 3D parity, structural deformation, or clinical validity.

The quadrature-backed cross-model velocity diagnostic answers where the selected 1D velocity field differs from the available resolved 3D velocity data at $t = 1.0$ s. For C23 and C40, the section-wise area-mean velocity errors have sample-average L_1 errors of 0.505 and 0.966 cm/s, root-mean-square L_2 errors of 0.664 and 1.87 cm/s, and L_∞ maxima of 2.35 and 9.92 cm/s. The largest discrepancies localize near the stenosis throat and in the reconstructed radial profiles, especially for C40. This supports mismatch localization under the declared tetrahedral cross-section quadrature operator, not a claim that the 1D model has been validated against resolved flow.

Pressure-ratio and shear-proxy notation are retained only as reduced model-output vocabulary for later work. The present manuscript does not add a pressure-drop result, a pressure-ratio result, a reported FFR or CT-FFR comparison, or an external measurement validation. A pressure-focused follow-up would need declared proximal and distal pressure observations, reference and averaging conventions, archived source identifiers or hashes, and a matched comparison record. A stronger velocity follow-up would also preserve per-run diagnostics such as timestep counts, CFL extrema, positivity checks, residual summaries, and closure/boundary perturbations varied one at a time.

The feasible local three-dimensional extension is a fixed-wall stationary-Stokes refinement study on the same smooth stenotic geometry. That study can report resolved velocity, pressure, and sampled wall-traction/WSS diagnostics. A full stenotic FSI study with wall displacement, wall velocity, structural stress, and interface work remains a separate future model tier requiring a structural solver and interface contract beyond the present implementation.

A Acronyms and Abbreviations

General abbreviations.

a.e.	almost everywhere	bpm	beats per minute
e.g.	*exempli gratia* (for example)	i.e.	*id est* (that is)

Clinical, modeling, and numerical acronyms.

CAS	coronary artery stenosis	FSI	fluid–structure interaction
CCTA	coronary computed tomography angiography	RBC	red blood cell
CT-FFR	computed tomography-derived fractional flow reserve	FFR	fractional flow reserve
IVUS	intravascular ultrasound	WSS	wall shear stress
OCT	optical coherence tomography	CFD	computational fluid dynamics
ODE	ordinary differential equation	PDE	partial differential equation
IC	initial condition	BC	boundary condition
FDM	finite difference method	FEM	finite element method
FVM	finite volume method	ROM	reduced-order model
0D	zero-dimensional	1D	one-dimensional
2D	two-dimensional	3D	three-dimensional

B Mathematical Notation

Notation B.1 (Report mathematical notation)**.** This appendix collects the mathematical notation used throughout this report.

Symbol	**Meaning**
$:=$	defines
$\equiv$	notational equivalence / identified with
$\mathbb{R}$	real numbers
$\mathbb{R}^+$	strictly positive real numbers
$\mathbb{R}^-$	strictly negative real numbers
$\mathbb{R}^n$	n-dimensional real vector space
$\mathbb{N}$	natural numbers
$\mathbb{N}_0$	nonnegative integers $\{0, 1, 2, \ldots\}$
$\{\boldsymbol{v_1}, \ldots, \boldsymbol{v_n}\}$	general basis of $\mathbb{R}^n$
$\{\boldsymbol{e_1}, \ldots, \boldsymbol{e_n}\}$	standard basis of $\mathbb{R}^n$
$[n]$	index set $\{1, 2, \ldots, n\} \subset \mathbb{N}$ for $n \in \mathbb{N}$
I_T	open time interval $(0, T_{\text{final}})$ for transient interior-time arguments
$\overline{I}_T$	closed time interval $[0, T_{\text{final}}]$ for initial-time data and flow maps
K	compact space-time rectangle, such as $[0, \ell] \times [0, T_{\text{final}}]$, used for output-field norms when locally stated
$\Omega \subset \mathbb{R}^n$	bounded domain
$\overline{\Omega}$	the closure of Ω
$\partial\Omega$	the boundary of Ω
$\Omega_B(t)$	spatial blood domain at time t
$\mathcal{Q}_T$	spatial-temporal domain $\{(t, \boldsymbol{x}) : t \in I_T,\ \boldsymbol{x} \in \Omega_B(t)\}$
$C^k(\Omega)$	functions with continuous classical derivatives through order k in Ω
$C^k(\overline{\Omega})$	functions whose derivatives through order k extend continuously to $\overline{\Omega}$
$C^k_c(\Omega)$	k times continuously differentiable functions with compact support in Ω
$C^\infty_c(\Omega)$	smooth functions with compact support in Ω

Symbol	Meaning
$C(K)$	continuous functions on a stated compact set K
$L^p(\Omega)$	Lebesgue space of p-integrable functions modulo equality a.e.
$d\boldsymbol{x}$	Lebesgue volume element on $\mathbb{R}^n$
$dS_{\boldsymbol{x}}$	surface measure element on $\partial\Omega \subset \mathbb{R}^n$
dV	3D volume element on domain $\Omega \subset \mathbb{R}^3$
dS	3D surface element on boundary $\partial\Omega \subset \mathbb{R}^3$
∇	gradient operator
$\partial_{x_i}\phi$	partial derivative of ϕ with respect to coordinate x_i
$\partial_t\phi$, $\partial_z f$, $\partial_A p$	compact partial derivatives with respect to named coordinates t, z, and A
$\partial_r u$, $\partial_\theta u$	compact partial derivatives in cylindrical coordinates
$\Delta\phi = \nabla \cdot \nabla\phi$	Laplacian of scalar field ϕ
div	divergence of a vector field
div	row-wise divergence of a tensor field
$\boldsymbol{v}_i$	i-th component of vector $\boldsymbol{v}$
$\boldsymbol{S} : \boldsymbol{T}$	Frobenius contraction $\sum_{i,j} S_{ij}T_{ij}$ for second-order tensors
$\langle\cdot,\ \cdot\rangle_X$	inner product on vector space X
$\langle\boldsymbol{u},\ \boldsymbol{v}\rangle \equiv \langle\boldsymbol{u},\ \boldsymbol{v}\rangle_{\mathbb{R}^n}$	inner product of vectors $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$
$\partial_{\hat{\boldsymbol{n}}}\phi$	normal derivative $\langle\nabla\phi,\ \hat{\boldsymbol{n}}\rangle$ of scalar field ϕ on $\partial\Omega$
$\Vert\cdot\Vert_X$	norm on a specified normed space X; bare $\Vert\cdot\Vert$ is local shorthand
$\Vert\cdot\Vert_{\mathrm{F}}$	Frobenius norm of a matrix or second-order tensor

C Domain Specific Notation

Definition C.1 (SI unit convention)**.** Unless otherwise stated, physical units are SI: length in meters, time in seconds, density in $\mathrm{kg/m^3}$, pressure in Pa, dynamic viscosity in $\mathrm{Pa \cdot s}$, and kinematic viscosity in $\mathrm{m^2/s}$.

Convention C.2 (Centimeter–second numerical-record convention)**.** The SI convention above governs continuum notation and unqualified analytical parameters. The Julia solver outputs and the Section 2 tables instead use the centimeter–gram–second convention of the numerical realization: lengths are in cm, time in s, density in $\mathrm{g/cm^3}$, velocity in cm/s, pressure and stress in $\mathrm{dyn/cm^2}$, dynamic viscosity in $\mathrm{g/(cm\,s)} = \mathrm{dyn\,s/cm^2}$, and kinematic viscosity in $\mathrm{cm^2/s}$. Thus the default solver geometry parameters L, $R_{\max}$, z_a, z_s, σ_a, and κ are centimeter quantities, while the stenosis localization λ in $\exp(-\lambda g^4)$ has units $\mathrm{cm^{-4}}$. The radius-squared solver state coordinate $a = R^2$ has units $\mathrm{cm^2}$; when a physical circular area is meant it is $A_{\text{phys}} = \pi a$, also reported in square centimeters.

Output fields carry their display units in their names. In particular, `z_cm` and `R0_cm` are geometry fields, `A_cm2` is the area-coordinate field in $\mathrm{cm^2}$, and `Q_cm3_s` is the flow-coordinate field paired with a in nominal $\mathrm{cm^3/s}$. For the radius-squared solver convention, the corresponding physical circular-area flow is $Q_{\text{phys}} = \pi Q$, while `uavg_cm_s` denotes $Q/a = Q_{\text{phys}}/A_{\text{phys}}$. The Section 2 velocity/error columns are in cm/s, `pressure_dyn_cm2` and initial-condition pressure-drop fields are in $\mathrm{dyn/cm^2}$, and `nu_eff_cm2_s` is in $\mathrm{cm^2/s}$. A pressure supplied through a Pa-valued command-line option is converted before storage using $1\,\mathrm{Pa} = 10\,\mathrm{dyn/cm^2}$. Any numerical-output value used in an SI formula must first be converted; for example $1\,\mathrm{cm} = 10^{-2}\,\mathrm{m}$, $1\,\mathrm{cm^2/s} = 10^{-4}\,\mathrm{m^2/s}$, $1\,\mathrm{g/(cm\,s)} = 10^{-1}\,\mathrm{Pa \cdot s}$, and $1\,\mathrm{dyn/cm^2} = 10^{-1}\,\mathrm{Pa}$.

Convention C.3 (Pressure-output convention)**.** Pressure-drop and pressure-ratio quantities are reduced output vocabulary. A pressure-drop trace requires proximal and distal scalar pressure observations, and a pressure-ratio output is reported only with a ratio-output specification that declares proximal and distal scalar pressure observations, a time-averaging window, a pressure reference, and the denominator admissibility condition below. The general ratio-output specification is

$$\pi_{\text{PR}} := (\mathcal{O}_{\text{prox}}, \mathcal{O}_{\text{dist}}, \mathcal{T}_{\text{HF}}, p_{\text{ref}}).$$

Here $\mathcal{O}_{\text{prox}}$ and $\mathcal{O}_{\text{dist}}$ are the declared scalar observation maps applied to the gauge-pressure field p_{g}, and p_{ref} is the fixed scalar pressure reference added to observed gauge-pressure traces before forming the ratio. In the 1D axial-tap or cross-sectional pressure-average specialization, this report may use the shorthand

$$\pi_{\text{PR}} = (z_{\text{prox}}, z_{\text{dist}}, \mathcal{T}_{\text{HF}}, p_{\text{ref}}),$$

with the pressure observation maps understood from the declared tap or cross-sectional averaging convention.

Given a model gauge-pressure field, the observed pressure traces are

$$P_{\text{prox}}(t) \ := \ \mathcal{O}_{\text{prox}}[p_{\text{g}}(\cdot,t)], \qquad\qquad P_{\text{dist}}(t) \ := \ \mathcal{O}_{\text{dist}}[p_{\text{g}}(\cdot,t)].$$

The corresponding pressure-drop trace is

$$\Delta p(t) \ := \ P_{\text{prox}}(t) - P_{\text{dist}}(t).$$

A pressure ratio is not invariant under common additive pressure shifts. The reference-consistent ratio traces are therefore defined after observation:

$$P^{\text{ratio}}_{\text{prox}}(t) \ := \ p_{\text{ref}} + P_{\text{prox}}(t), \qquad\qquad P^{\text{ratio}}_{\text{dist}}(t) \ := \ p_{\text{ref}} + P_{\text{dist}}(t).$$

This trace-level convention does not require the observation maps to preserve constant shifts. The subscript HF labels a declared high-flow averaging window in the study record; it is not a clinical hyperemia model unless an external measurement comparison is separately specified [18, Ch. 5, Sec. 5.5, pp. 72–73] [21, p. 1703]. The ratio-output specification is admissible for a computed pressure trace when $\mathcal{T}_{\text{HF}} \subset I_T$ is Lebesgue-measurable with $0 < |\mathcal{T}_{\text{HF}}| < \infty$, the proximal and distal ratio-pressure traces are integrable on that window, and

$$\langle P^{\text{ratio}}_{\text{prox}}(t)\rangle_{\mathcal{T}_{\text{HF}}} \neq 0.$$

For an admissible π_{PR}, the corresponding reduced pressure-ratio output is

$$\text{PR}_{1D}(\pi_{\text{PR}}) \ := \ \frac{\langle P^{\text{ratio}}_{\text{dist}}(t)\rangle_{\mathcal{T}_{\text{HF}}}}{\langle P^{\text{ratio}}_{\text{prox}}(t)\rangle_{\mathcal{T}_{\text{HF}}}}.$$

For an integrable scalar trace,

$$\langle f\rangle_{\mathcal{T}_{\text{HF}}} \ := \ \frac{1}{|\mathcal{T}_{\text{HF}}|}\int_{\mathcal{T}_{\text{HF}}} f(t)\,dt.$$

This bracket denotes time averaging, not the inner product notation in Appendix B. An FFR-like shorthand refers to this reduced ratio only for records carrying an admissible π_{PR}; the shorthand does not add a clinical measurement model [18, Ch. 5, Sec. 5.5, pp. 72–73] [21, p. 1703]. If π_{PR} is absent or inadmissible, the pressure ratio is outside the reported output space.

Convention C.4 (Shear-output convention)**.** Shear terminology is model-tier dependent. A resolved wall shear stress output requires a resolved stress tensor, wall normal, wall location, output operator, and, for a signed component, a chosen tangent direction [16, Abstract, Sec. 2, pp. 8–10]. A reduced shear proxy such as $\tau^{1D}_{w,z}$ requires the model tier, wall closure, rheology descriptor, velocity-profile or friction closure, evaluation location or spatial averaging operator, and sign or time-window convention. Reduced shear proxies are closure diagnostics; they are not interchangeable with resolved traction-based wall shear stress unless a separate comparison map is declared.

Symbol	Meaning	Units
R	vessel radius; vessel diameter is $d_v := 2R$	m
d_v	vessel diameter	m
η	dynamic viscosity	$\mathrm{Pa} \cdot \mathrm{s}$
$\nu := \eta/\rho$	kinematic viscosity	m^2/s
$\eta_{\text{eff}}(\dot{\gamma})$	effective dynamic viscosity for generalized Newtonian models	$\mathrm{Pa} \cdot \mathrm{s}$
$\nu_{\text{eff}}(\dot{\gamma}) := \eta_{\text{eff}}(\dot{\gamma})/\rho$	effective kinematic viscosity after density scaling	m^2/s
τ	shear stress	Pa
$\dot{\gamma}$	shear rate	s^{-1}
ρ	density field	kg/m^3
p	pressure field	Pa
$\boldsymbol{u}$	velocity field	m/s
$D(\boldsymbol{u})$	rate-of-deformation tensor, $D(\boldsymbol{u}) := \frac{1}{2}\left(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^\top\right)$	s^{-1}
$W_0 := d_v\sqrt{\frac{\omega\rho}{\eta}}$	Womersley number, diameter-based convention	$-$
Re	Reynolds number	$-$
t	time	s
T or T_{final}	terminal time; use T_{final} where it could be confused with stress tensors	s
ω	angular frequency	rad/s
(r, θ, z)	cylindrical coordinates used only when 3D coordinate forms are written	$\mathrm{m}, -, \mathrm{m}$
$(\boldsymbol{e}_r, \boldsymbol{e}_\theta, \boldsymbol{e}_z)$	cylindrical orthonormal basis vectors	$-$
τ_{ij}	cylindrical viscous-stress component	Pa
$\boldsymbol{f_b}$	body force per unit mass	m/s^2
$\rho\boldsymbol{f_b}$	body force per unit volume	N/m^3

Reduced-model and forward-map notation.

Symbol	Meaning	Units
$A(z,t)$	cross-sectional lumen area in the 1D model	m^2
$Q(z,t)$	volumetric flow rate through a cross-section	m^3/s
$\bar{u}(z,t) := Q/A$	section-mean axial velocity	m/s
$I_z = (0,L)$	axial single-vessel interval for a 1D formulation	m
$\mathbf{U} = (A,Q)^\top$	conservative area-flow state vector	$(\mathrm{m}^2, \mathrm{m}^3/\mathrm{s})$
$\mathcal{P}$	cross-sectional velocity-profile descriptor used by the 1D closure; admitted values include flat, parabolic, and power-family profiles	mixed
$\alpha_{\mathcal{P}}$	momentum-flux correction induced by the declared profile $\mathcal{P}$; the parabolic baseline uses $\alpha_{\mathcal{P}} = 4/3$	–
γ	power-profile exponent in $u(r) = ((\gamma+2)/\gamma)\bar{u}(1-(r/R)^\gamma)$	–
$g_{\mathcal{P}}$	profile shear-rate factor used by the 1D rheology proxy; for power profiles $g_{\mathcal{P}} = \gamma + 2$, while flat profiles assign this value explicitly	–
$R(A(z,t)) = \sqrt{A(z,t)/\pi}$	current reduced lumen radius derived from the 1D area	m
$R_0(z)$	reference stenosed radius used by the wall law	m
$R_{\mathrm{base}}(z)$	nominal healthy baseline radius before applying $s(z)$	m
$\mathcal{R}$	rheology descriptor in the 1D model; distinct from radius symbols R, R_0, and R_{base}; admitted values are `newtonian`, `carreau`, `carreau-yasuda`, `casson`, and `power-law`	mixed
$\mathcal{W}$	wall descriptor; the selected 1D report model uses the Canic–Koiter member of $\mathcal{W}_{\mathrm{elastic1D}}$, while fixed-wall and resolved-FSI wall models require separate declarations	mixed
$\mathcal{C}$	coupling/interface descriptor; the selected single-vessel study uses $\mathcal{C}_{\mathrm{isolated}}$	mixed
$s(z)$	dimensionless stenosis profile, with $0 \le s(z) < 1$	–
$A_0(z) = \pi R_0(z)^2$	wall-law reference area after applying the stenosis profile	m^2
$s_{\max}$	maximum fractional radius-reduction parameter in $s(z)$	–
$g(z)$	smooth asymmetric stenosis-kernel coordinate	m
$\psi(z) = \exp(-\lambda g(z)^4)$	smooth rapidly localized stenosis kernel used by the baseline profile $s(z) = s_{\max}\psi(z)$	–

Symbol	Meaning	Units
z_a	center of the Gaussian asymmetry term in $g(z)$	m
z_s	axial shift parameter in $g(z)$	m
σ_a	width parameter for the Gaussian asymmetry term in $g(z)$	m
κ	amplitude of the Gaussian asymmetry term in $g(z)$	m
λ	localization strength in $\psi(z) = \exp(-\lambda g(z)^4)$	m^{-4}
$\beta(z)$	wall-stiffness parameter in $p_{\mathrm{g}} = p_{\mathrm{ext}} + \beta A_0^{-1}(\sqrt{A} - \sqrt{A_0})$	$\mathrm{Pa} \cdot \mathrm{m}$
$p_{\mathrm{ext}}(z,t)$	external-pressure value in the wall-law gauge convention; baseline value $p_{\mathrm{ext}} \equiv 0$ unless declared otherwise	Pa
$\Psi(A,z,t)$ or $\Psi_{\mathcal{W}}(A,z,t)$	elastic flux potential satisfying $\partial_A \Psi_{\mathcal{W}} = (A/\rho)\partial_A p_{\mathrm{g}}^{\mathcal{W}}$ and normalized by $\Psi_{\mathcal{W}}(A_0(z),z,t) = 0$ when that normalization is declared	$\mathrm{m}^4/\mathrm{s}^2$
$\mathbf{F}$	area-flow flux map; not a residual or objective functional	problem-dependent
$\mathbf{S}$	source-term map in the 1D balance-law template	problem-dependent
$S_{\mathrm{geom}}^{\mathcal{W}}$	fixed-area wall/geometry source associated with the selected reduced wall closure $\mathcal{W}$	$\mathrm{m}^3/\mathrm{s}^2$
$S_f^{\mathcal{R}}$	wall-friction source associated with the selected rheology or effective-viscosity closure $\mathcal{R}$	$\mathrm{m}^3/\mathrm{s}^2$
$\widehat{\mathbf{F}}_{i+1/2}^n$	numerical flux at a finite-volume cell interface and time level	problem-dependent
$\widehat{\mathbf{S}}_i^n$	cell source quadrature used by a computed finite-volume scheme	problem-dependent
$\Delta z_i, \Delta t^n$	finite-volume cell length and timestep in a computed solve	m, s
$\dot{\gamma}_{1D}$	computed shear-rate proxy $g_{\mathcal{P}}\lvert Q\rvert/(a\sqrt{a})$, where $a = R^2$	s^{-1}
$\nu_{\mathrm{eff}}^{\mathcal{R}}$	closure-indexed effective kinematic viscosity used in the reduced viscous/source and pressure-correction terms	m^2/s
C_f or $C_f(z,A)$	wall-friction coefficient in the 1D source $-C_f Q/A$; constant-viscosity value $C_f = 8\pi\eta/\rho$	m^2/s
$\eta_{\mathrm{fric}}(z,t)$	dynamic viscosity used by the reduced wall-friction and signed-shear-proxy formula; $\eta_{\mathrm{fric}} = \eta$ for constant-viscosity realizations and $\eta_{\mathrm{fric}} = \eta_{\mathrm{eff}}^{\mathcal{R}}(\dot{\gamma}_{1D})$ for admitted non-Newtonian descriptor realizations	$\mathrm{Pa} \cdot \mathrm{s}$

Symbol	Meaning	Units
c_{wave} or c	1D pulse-wave speed, $c^2 = (A/\rho)\partial_A p_{\mathrm{g}}$ when no concentration field is in scope	m/s
$e \in \mathcal{E}$	expert/closure record for a closure-indexed 1D forward-map family; suppress e only for single-expert baseline records	mixed
$\mathcal{G}_{1D,h}^{r_h,e}$	computed fixed-grid 1D map indexed by numerical record r_h and expert record e	mixed

Symbol	Meaning	Units
p_{3D}	continuum pressure in the 3D/NS setting	Pa
$p_{3D,\mathrm{g}}$	continuum pressure after subtracting the wall-law gauge reference	Pa
p_{g}	1D gauge pressure produced by the wall law	Pa
$p_{\mathrm{ratio}} := p_{\mathrm{ref}} + p_{\mathrm{g}}$	field-level reference-consistent pressure shorthand; ratio traces are defined after scalar observation by Convention C.3	Pa
p_{ref}	fixed pressure reference recorded by the ratio-output specification	Pa
$\mathcal{O}_{\mathrm{prox}}, \mathcal{O}_{\mathrm{dist}}$	proximal and distal scalar pressure observation maps declared for pressure-drop or pressure-ratio outputs	input/output Pa
$P_{\mathrm{prox}}, P_{\mathrm{dist}}$	observed proximal and distal gauge-pressure traces after applying the declared pressure observation maps	Pa
$P^{\mathrm{ratio}}_{\mathrm{prox}}, P^{\mathrm{ratio}}_{\mathrm{dist}}$	observed proximal and distal pressure traces after adding p_{ref}	Pa
$\pi_{\Delta p} = (z_{\mathrm{prox}}, z_{\mathrm{dist}})$	1D shorthand pressure-drop tap specification; the observation maps define the actual scalar traces	m, m
π_{PR}	ratio-output specification $(\mathcal{O}_{\mathrm{prox}}, \mathcal{O}_{\mathrm{dist}}, \mathcal{T}_{\mathrm{HF}}, p_{\mathrm{ref}})$; axial taps may be specified by shorthand	mixed
$\langle f \rangle_{\mathcal{T}_{\mathrm{HF}}}$	time average of an integrable trace over the declared high-flow window; not the Appendix B inner product	same as f
$\langle P^{\mathrm{ratio}}_{\mathrm{prox}} \rangle_{\mathcal{T}_{\mathrm{HF}}} \neq 0$	denominator admissibility condition required before PR_{1D} is reported	–
$p_{\mathrm{out,g}}$	prescribed outlet gauge pressure in the same reference frame as p_{g}	Pa
$\Delta p(t)$	proximal-minus-distal scalar pressure trace after the declared observations	Pa
$\overline{\Delta p}_{\mathcal{T}_{\mathrm{HF}}}$	scalar pressure-drop output averaged over the declared high-flow window $\mathcal{T}_{\mathrm{HF}}$	Pa
FFR-like shorthand	optional shorthand for PR_{1D} only when Convention C.3 and admissibility requirements are met	–
$\mathrm{PR}_{1D}(\pi_{\mathrm{PR}})$	reduced pressure-ratio output under the declared observation maps, pressure reference, and averaging window	–
τ_w or $\lvert\tau_w\rvert$	resolved traction-based wall-shear-stress output after declaring the stress tensor, wall normal, location, and output operator	Pa
$\tau^{1D}_{w,z}$	signed 1D axial shear proxy after declaring the model tier, closures, location, and sign or window convention	Pa
`shear_rate_1_s`	CSV output field for the computed shear-rate proxy $\dot{\gamma}_{1D}$	s^{-1}
`nu_eff_cm2_s`	CSV output field for the effective kinematic viscosity used by the declared rheology closure	cm^2/s
`rheology`	CSV and summary output field naming the declared rheology descriptor	–

Remark C.5 (Viscosity notation). We write η for dynamic viscosity and

$$\nu := \frac{\eta}{\rho}$$

for kinematic viscosity. The distinction is dimensional: this report follows the convention that the coefficient in the Cauchy stress tensor is a dynamic viscosity: $D(\boldsymbol{u})$ has units of inverse time, so $\eta D(\boldsymbol{u})$ has pressure units. For generalized Newtonian models, the scalar viscosity function multiplying $2D(\boldsymbol{u})$ in the extra stress is likewise an effective dynamic viscosity, denoted

$$\eta_{\text{eff}}(\dot{\gamma}), \qquad \dot{\gamma} \ := \ \sqrt{2\boldsymbol{D}(\boldsymbol{u}) : \boldsymbol{D}(\boldsymbol{u})}.$$

Here "this quantity" means the stress-level viscosity coefficient, not the divided-by-density coefficient. Several sources write the generalized viscosity function as $\mu(|D|^2)$, while related hemodynamics sources also use μ for viscosity coefficients in reduced and continuum settings [1, Sec. 3.1, p. 35, Sec. 3.6, p. 215] [16, Sec. 1.1, pp. 6–7, Eqs. (1.2)–(1.4), Fig. 4.4, p. 18]. To keep this report unambiguous, ν is always kinematic viscosity, while η and η_{eff} are dynamic viscosities.

D Mathematical Foundations

This appendix supplies citation anchors for the analytic notation used in the main chapters and in Appendix B. It separates domain and boundary language, classical function spaces, field notation, differential operators, and norm conventions so later sections can cite only the convention they need.

Convention D.1 (Standing regularity convention)**.** Unless a section states a stronger or weaker hypothesis, classical identities are used only under the regularity needed for their terms to be defined. In particular, fields have the displayed classical derivatives in the indicated domains, boundary traces are defined when boundary terms are written, and integration-by-parts or adjoint identities are invoked only when the required boundary terms are meaningful. When a weak formulation is discussed, the relevant Sobolev space, trace theorem, and boundary regularity assumptions are stated explicitly; Sobolev traces are not inferred from this classical convention. At minimum, boundaries are assumed piecewise C^1 when classical boundary integrals or integration-by-parts identities are used. Outward unit normals are then interpreted on the regular boundary pieces, hence surface-a.e. When a pointwise normal derivative is asserted on all of $\partial\Omega$, the boundary is assumed at least C^1.

Definition D.2 (Domain and boundary notation)**.** Let $\Omega \subset \mathbb{R}^n$ be a bounded open connected set, with $n \in \mathbb{N}$, and denote its closure by $\overline{\Omega}$. In the hemodynamics settings considered in this report, only $n \in [3]$ is used. The standing regularity convention in Convention D.1 applies whenever traces, outward unit normals, or integration by parts are used. For open Ω,

$$\overline{\Omega} = \Omega \cup \partial\Omega, \qquad \partial\Omega = \overline{\Omega} \setminus \Omega.$$

Since Ω is bounded, $\overline{\Omega}$ is closed and bounded and is therefore compact by the Heine–Borel theorem. The open set Ω itself need not be compact. Since Ω is bounded, $\partial\Omega$ is also compact. When the standing regularity convention supplies a piecewise C^1 boundary, the outward unit normal $\hat{\boldsymbol{n}}$ is interpreted on the regular boundary pieces, hence $dS_{\boldsymbol{x}}$-a.e. on $\partial\Omega$.

Definition D.3 (Time intervals and compact space-time sets)**.** For a final time $T_{\text{final}} > 0$, this report uses

$$I_T \;:=\; (0, T_{\text{final}}), \qquad \overline{I}_T \;:=\; [0, T_{\text{final}}].$$

The open interval I_T is used for interior-time differential identities and space-time PDE regions, while $\overline{I}_T$ is used when initial-time data, endpoint data, sampled time grids, or flow maps are included. When a local final time such as T_* is stated, the same convention applies with T_{final} replaced by that local final time.

For one-dimensional output fields, a compact space-time rectangle is written as

$$K \;:=\; [0, \ell] \times [0, T_{\text{final}}]$$

or by an explicitly stated local variant. Spaces such as $C(K)$ and $C^1([0,\ell])$ mean continuous functions on the named compact set, with the supremum norm when a norm is written. Spaces such as $L^p(K)$ use the product Lebesgue measure and the same almost-everywhere convention as Definition D.7.

Definition D.4 (Field and multi-index notation)**.** For $\Omega \subset \mathbb{R}^n$, we use the following field notation:

$$\begin{aligned}
&\phi : \Omega \to \mathbb{R} \;\text{ is a } \textit{scalar field}, \\
&\boldsymbol{f} : \Omega \to \mathbb{R}^n \;\text{ is a } \textit{vector field}, \\
&\boldsymbol{T} : \Omega \to \mathbb{R}^{n\times n} \;\text{ is a } \textit{(second-order) tensor field},
\end{aligned}$$

and we write $\phi(\boldsymbol{x})$, $\boldsymbol{f}(\boldsymbol{x})$, and $\boldsymbol{T}(\boldsymbol{x})$[1] for the values at a point $\boldsymbol{x} \in \Omega$. When referring to a physical quantity, its measurement units are indicated as $\left[\;\cdot\;\right]$, whereas dimensionless quantities are indicated as $\left[\;-\;\right]$. The k-th partial derivative of ϕ with respect to coordinate x_i is

$$\partial^k_{x_i}\phi \;\equiv\; \frac{\partial^k \phi}{\partial x_i^k}.$$

For a multi-index $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}_0^n$, set $|\alpha| = \alpha_1 + \cdots + \alpha_n$. When the derivative exists classically, the corresponding derivative of a scalar field ϕ is

$$D^\alpha \phi \;\equiv\; \frac{\partial^{|\alpha|}\phi}{\partial x_1^{\alpha_1}\,\partial x_2^{\alpha_2}\,\ldots\,\partial x_n^{\alpha_n}}.$$

For $|\alpha| = 1$, $D^\alpha\phi$ is a first-order partial derivative of ϕ, e.g. $\partial_{x_i}\phi$ where $\alpha_i = 1$ and $\alpha_j = 0$ for all $j \in [n]\setminus\{i\}$.

Definition D.5 (Classical function-space notation)**.** Given a domain Ω and a scalar field $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, a scalar function space on Ω is an $\mathbb{F}$-vector subspace of the maps $f : \Omega \to \mathbb{F}$. Unless a complex-valued space is explicitly specified, this report uses real-valued spaces. Thus $C^0(\Omega)$ denotes the real-valued continuous functions on the open domain Ω. For $k \in \mathbb{N}_0$, $C^k(\Omega)$ denotes functions $\phi : \Omega \to \mathbb{R}$ such that $D^\alpha\phi$ exists and is continuous in Ω for every $|\alpha| \le k$. By contrast, $C^k(\overline{\Omega})$ denotes functions in $C^k(\Omega)$ whose derivatives through order k extend continuously to the closure $\overline{\Omega}$. The space $C^\infty(\Omega)$ is the intersection of all $C^k(\Omega)$ for $k \in \mathbb{N}_0$, i.e. the infinitely differentiable functions on Ω. The notation $C^k(\Omega; \mathbb{R}^m)$ denotes the maps $\boldsymbol{f} : \Omega \to \mathbb{R}^m$ with components in $C^k(\Omega)$, i.e., $\boldsymbol{f} = (f_i)_{i\in[m]}$ with $f_i \in C^k(\Omega)$ for all $i \in [m]$.

Definition D.6 (Differential-operator conventions)**.** For real-valued $\phi \in C^1(\Omega)$, the partial derivative with respect to coordinate x_i is

$$\partial_{x_i}\phi = \frac{\partial\phi}{\partial x_i}.$$

[1] A second-order tensor value $\boldsymbol{T}(\boldsymbol{x}) \in \mathbb{R}^{n\times n}$ can be identified with the matrix of a linear map $\mathbb{R}^n \to \mathbb{R}^n$ acting on vectors by matrix-vector multiplication.

Named-coordinate partial derivatives use the same compact convention; for example,

$$\partial_t \phi = \frac{\partial \phi}{\partial t}, \qquad \partial_z f = \frac{\partial f}{\partial z}, \qquad \partial_A p = \frac{\partial p}{\partial A}.$$

Cylindrical derivatives such as $\partial_r u$ and $\partial_\theta u$ are read analogously. Boundary notation retains ∂, as in $\partial V(t)$ and $\partial\Omega$. For $\boldsymbol{x} \in \Omega$ and a direction $\boldsymbol{v} \in \mathbb{R}^n$, the directional derivative is

$$D_{\boldsymbol{v}}\phi(\boldsymbol{x}) \;:=\; \langle \nabla\phi(\boldsymbol{x}),\; \boldsymbol{v}\rangle_{\mathbb{R}^n}.$$

If $\phi \in C^1(\overline{\Omega})$, $\boldsymbol{x} \in \partial\Omega$, and $\hat{\boldsymbol{n}}$ is the outward unit normal at $\boldsymbol{x}$, then the boundary normal derivative is

$$\partial_{\hat{\boldsymbol{n}}}\phi(\boldsymbol{x}) \;:=\; \langle \nabla\phi(\boldsymbol{x}),\; \hat{\boldsymbol{n}}\rangle_{\mathbb{R}^n}.$$

For Lebesgue measure $d\boldsymbol{x}$ on $\mathbb{R}^n$ the integral of ϕ in Ω is

$$\int_\Omega \phi(\boldsymbol{x})\, d\boldsymbol{x}.$$

If $\partial\Omega$ is of class C^1 and $\phi \in C^0(\overline{\Omega})$, then the classical boundary trace $\phi|_{\partial\Omega}$ is continuous on $\partial\Omega$. Since Ω is bounded and $\partial\Omega$ is a compact C^1 hypersurface, this trace belongs to $L^1(\partial\Omega; dS_{\boldsymbol{x}})$. The corresponding boundary integral is written

$$\int_{\partial\Omega} \phi(\boldsymbol{x})\; dS_{\boldsymbol{x}}.$$

We use the column-gradient convention for scalar fields. For $\phi \in C^1(\Omega)$, the gradient is

$$\nabla\phi \;:=\; (\partial_{x_i}\phi)_{i\in[n]} \in \mathbb{R}^n.$$

For $\phi \in C^2(\Omega)$, the Laplacian is

$$\Delta\phi \;:=\; \nabla\cdot\nabla\phi \;=\; \sum_{i\in[n]} \partial^2_{x_i}\phi.$$

For vector-valued $\boldsymbol{f} \in C^1(\Omega;\mathbb{R}^n)$ with components $\boldsymbol{f} = (f_i)_{i\in[n]}$, the vector gradient is the Jacobian with entries $(\nabla\boldsymbol{f})_{ij} \;:=\; \partial_{x_j} f_i$. The divergence is the trace of this Jacobian:

$$\operatorname{div}\boldsymbol{f} \;:=\; \nabla\cdot\boldsymbol{f} \;=\; \sum_{i\in[n]} \partial_{x_i} f_i \;=\; \operatorname{tr}(\nabla\boldsymbol{f}).$$

For $\boldsymbol{f} \in C^2(\Omega;\mathbb{R}^n)$, the Laplacian is defined componentwise. The domain integral is also componentwise whenever the components are integrable:

$$\begin{aligned}
\Delta \boldsymbol{f} &:= (\Delta f_i)_{i\in[n]} \\
\int_\Omega \boldsymbol{f}(\boldsymbol{x})\, d\boldsymbol{x} &:= \left(\int_\Omega f_i(\boldsymbol{x})\, d\boldsymbol{x}\right)_{i\in[n]}.
\end{aligned}$$

For tensor-valued $\boldsymbol{T} \in C^1(\Omega;\mathbb{R}^{n\times n})$ with entries $\boldsymbol{T} = (T_{ij})_{i,j\in[n]}$, we use the row-divergence convention $(\mathbf{div}\boldsymbol{T})_i = \sum_{j\in[n]} \partial_{x_j} T_{ij}$. Equivalently, if $\boldsymbol{t}_i = (T_{ij})_{j\in[n]}$ denotes the i-th row of $\boldsymbol{T}$, then

$$\begin{aligned}
\mathbf{div}\,\boldsymbol{T} &:= (\operatorname{div} \boldsymbol{t}_i)_{i\in[n]} \\
&= (\nabla\cdot \boldsymbol{t}_i)_{i\in[n]} \\
&= \left(\sum_{j\in[n]} \partial_{x_j} T_{ij}\right)_{i\in[n]}.
\end{aligned}$$

For $\boldsymbol{T} \in C^2(\Omega;\mathbb{R}^{n\times n})$, the Laplacian is defined componentwise. The domain integral is also defined componentwise whenever the entries are integrable:

$$\begin{aligned}
\Delta \boldsymbol{T} &:= (\Delta T_{ij})_{i,j\in[n]} \\
\int_\Omega \boldsymbol{T}(\boldsymbol{x})\, d\boldsymbol{x} &:= \left(\int_\Omega T_{ij}(\boldsymbol{x})\, d\boldsymbol{x}\right)_{i,j\in[n]}.
\end{aligned}$$

Definition D.7 (C^k and L^p norm conventions)**.** Let $\Omega \subset \mathbb{R}^n$ be bounded and open. We write $C^k(\overline{\Omega})$ for the functions $\phi \in C^k(\Omega)$ such that each derivative $D^\alpha\phi$, $|\alpha| \le k$, extends continuously to $\overline{\Omega}$. The norm is

$$\|\phi\|_{C^k(\overline{\Omega})} := \sum_{|\alpha|\le k} \sup_{\boldsymbol{x}\in\overline{\Omega}} |D^\alpha \phi(\boldsymbol{x})|.$$

With this norm, $C^k(\overline{\Omega})$ is a Banach space: every Cauchy sequence in $C^k(\overline{\Omega})$ converges, with respect to $\|\cdot\|_{C^k(\overline{\Omega})}$, to a limit in $C^k(\overline{\Omega})$. In particular, for $k = 0$, we have

$$\|\phi\|_{C^0(\overline{\Omega})} := \sup_{\boldsymbol{x}\in\overline{\Omega}} |\phi(\boldsymbol{x})| = \max_{\boldsymbol{x}\in\overline{\Omega}} |\phi(\boldsymbol{x})|.$$

For continuous functions on compact sets, the max norm agrees with the usual supremum norm.

For $1 \le p < \infty$, $L^p(\Omega)$ denotes the space of equivalence classes of real-valued measurable functions on Ω, identified up to equality almost everywhere, with finite norm

$$\|\phi\|_{L^p(\Omega)} := \left(\int_\Omega |\phi(\boldsymbol{x})|^p\, d\boldsymbol{x}\right)^{1/p}.$$

For $p = \infty$,

$$\|\phi\|_{L^\infty(\Omega)} \;:=\; \operatorname*{ess\,sup}_{\boldsymbol{x}\in\Omega} |\phi(\boldsymbol{x})|.$$

With this almost-everywhere identification, $L^p(\Omega)$ is a Banach space for $1 \leq p \leq \infty$.

The notation $L^p(\Omega; \mathbb{R}^m)$ denotes equivalence classes of measurable maps $\boldsymbol{f} : \Omega \to \mathbb{R}^m$, identified up to equality a.e., whose components belong to $L^p(\Omega)$, i.e., $\boldsymbol{f} = (f_i)_{i\in[m]}$ with $f_i \in L^p(\Omega)$ for all $i \in [m]$. When a norm on $L^p(\Omega; \mathbb{R}^m)$ is needed, we use the Euclidean norm in the range. For $1 \leq p < \infty$,

$$\|\boldsymbol{f}\|_{L^p(\Omega;\mathbb{R}^m)} \;:=\; \left(\int_\Omega \|\boldsymbol{f}(\boldsymbol{x})\|_{\mathbb{R}^m}^p \, d\boldsymbol{x}\right)^{1/p}.$$

For $p = \infty$, the corresponding convention is

$$\|\boldsymbol{f}\|_{L^\infty(\Omega;\mathbb{R}^m)} \;:=\; \operatorname*{ess\,sup}_{\boldsymbol{x}\in\Omega} \|\boldsymbol{f}(\boldsymbol{x})\|_{\mathbb{R}^m}.$$

For matrix or second-order tensor values, the pointwise range norm is the Frobenius norm

$$\|\boldsymbol{T}\|_{\mathrm{F}} \;:=\; (\boldsymbol{T} : \boldsymbol{T})^{1/2} = \left(\sum_{i,j\in[n]} T_{ij}^2\right)^{1/2}.$$

Thus tensor-valued L^p norms are formed by replacing $|\phi(\boldsymbol{x})|$ in the scalar L^p formula with $\|\boldsymbol{T}(\boldsymbol{x})\|_{\mathrm{F}}$.

E Continuum Derivation Details

This appendix supplements the continuum mechanics details that support the main-body assumptions without interrupting the reduced-order forward-model narrative within this report. It is appendix background for notation and model hierarchy, not additional evidence for the numerical study. All identities below are classical identities under the regularity needed for the displayed derivatives, traces, and changes of variables. The standing regularity convention, boundary notation, and differential-operator convention are those of Convention D.1, Definition D.2, and Definition D.6. In particular, **div** is the row-divergence of a tensor field, and boundary forms assume the stated trace and normal hypotheses; no weak formulation is being asserted in this appendix.

E.1 Material Derivative and Trajectory Details

For a scalar field $\phi \in C^1(\mathcal{Q}_T)$ and a sufficiently regular flow map $\mathcal{L}_t$, define the Lagrangian pullback $\widehat{\phi}(t,\boldsymbol{\xi}) := \phi(t,\mathcal{L}_t(\boldsymbol{\xi}))$. Differentiating with respect to time along a trajectory and using $\partial_t\mathcal{L}_t(\boldsymbol{\xi}) = \boldsymbol{u}(t,\mathcal{L}_t(\boldsymbol{\xi}))$ gives

$$\frac{d}{dt}\phi(t,\mathcal{L}_t(\boldsymbol{\xi})) = \partial_t\phi(t,\boldsymbol{x}) + \nabla\phi(t,\boldsymbol{x})\cdot\boldsymbol{u}(t,\boldsymbol{x}), \qquad \boldsymbol{x} = \mathcal{L}_t(\boldsymbol{\xi}).$$

The identity

$$\mathrm{div}(\phi\boldsymbol{u}) = \nabla\phi\cdot\boldsymbol{u} + \phi\mathrm{div}\boldsymbol{u}$$

also gives

$$D_t\phi = \partial_t\phi + \mathrm{div}(\phi\boldsymbol{u}) - \phi\mathrm{div}\boldsymbol{u}.$$

E.2 Jacobian Evolution and Reynolds Transport

Let $\widehat{\boldsymbol{F}}_t = \nabla_{\boldsymbol{\xi}}\mathcal{L}_t$ and $\widehat{J}_t = \det\widehat{\boldsymbol{F}}_t$. Here $\mathcal{L}_t$ is assumed to be an orientation-preserving C^1 change of variables on the material volume being considered, with the additional time regularity needed to differentiate $\widehat{\boldsymbol{F}}_t$ and $\widehat{J}_t$. The standard Jacobi identity gives

$$\partial_t\widehat{J}_t = \widehat{J}_t\,\mathrm{tr}\Big(\partial_t\widehat{\boldsymbol{F}}_t\,\widehat{\boldsymbol{F}}_t^{-1}\Big) = \widehat{J}_t\,\mathrm{div}\boldsymbol{u}(t,\mathcal{L}_t(\boldsymbol{\xi})),$$

which is the Lagrangian form of $D_tJ_t = J_t\mathrm{div}\boldsymbol{u}$ [2, Part I, Ch. 1, Sec. 1.3].

For $V(t) = \mathcal{L}_t(V(0))$, change variables in $I(t) = \int_{V(t)} \phi(t, \boldsymbol{x})\, d\boldsymbol{x}$:

$$I(t) = \int_{V(0)} \phi(t, \mathcal{L}_t(\boldsymbol{\xi}))\widehat{J}_t(\boldsymbol{\xi})\, d\boldsymbol{\xi}.$$

Differentiating under the integral sign and substituting the Jacobian-evolution identity gives

$$I'(t) = \int_{V(t)} (D_t\phi + \phi \mathrm{div}\boldsymbol{u})\ d\boldsymbol{x} = \int_{V(t)} (\partial_t\phi + \mathrm{div}(\phi\boldsymbol{u}))\ d\boldsymbol{x}.$$

If $V(t)$ has Lipschitz boundary and the trace of $\phi\boldsymbol{u}$ is integrable on $\partial V(t)$, the divergence theorem gives the boundary form

$$I'(t) = \int_{V(t)} \partial_t\phi\ d\boldsymbol{x} + \int_{\partial V(t)} \phi\, \boldsymbol{u} \cdot \hat{\boldsymbol{n}}\, dS.$$

E.3 Linear Momentum Balance Details

Starting from the material-volume balance

$$\frac{d}{dt}\int_{V(t)} \rho\boldsymbol{u}\, dV = \int_{\partial V(t)} \boldsymbol{T}\hat{\boldsymbol{n}}\, dS + \int_{V(t)} \rho \boldsymbol{f_b}\, dV,$$

Reynolds transport gives

$$\frac{d}{dt}\int_{V(t)} \rho\boldsymbol{u}\, dV = \int_{V(t)} (D_t(\rho\boldsymbol{u}) + \rho\boldsymbol{u}\, \mathrm{div}\boldsymbol{u})\, dV.$$

Using mass conservation, $D_t\rho + \rho\mathrm{div}\boldsymbol{u} = 0$, this reduces to

$$\int_{V(t)} \rho D_t\boldsymbol{u}\, dV = \int_{V(t)} \rho\, (\partial_t\boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u})\, dV.$$

The surface traction term satisfies

$$\int_{\partial V(t)} \boldsymbol{T}\hat{\boldsymbol{n}}\, dS = \int_{V(t)} \mathbf{div}(\boldsymbol{T})\, dV.$$

This is the tensor row-divergence convention from Definition D.6, applied under the regularity needed for the stress trace $\boldsymbol{T}\hat{\boldsymbol{n}}$ and the volume integral of $\mathbf{div}\boldsymbol{T}$. Since $V(t)$ is arbitrary within the localization class, the differential form is

$$\rho\, (\partial_t\boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}) = \mathbf{div}(\boldsymbol{T}) + \rho\boldsymbol{f_b}.$$

The conservative Eulerian form

$$\partial_t(\rho\boldsymbol{u}) + \mathbf{div}(\rho\boldsymbol{u}\otimes\boldsymbol{u}) = \mathbf{div}(\boldsymbol{T}) + \rho\boldsymbol{f_b}$$

is equivalent whenever the continuity equation holds, because

$$\partial_t(\rho\boldsymbol{u}) + \mathbf{div}(\rho\boldsymbol{u}\otimes\boldsymbol{u}) = \rho\left(\partial_t\boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) + \boldsymbol{u}\left(\partial_t\rho + \operatorname{div}(\rho\boldsymbol{u})\right).$$

F Navier–Stokes Coordinate and Energy Details

This appendix collects mathematical orientation material for the 3D Navier–Stokes model. It is background for the continuum hierarchy; it is not evidence for the 1D study specification or any downstream validation claim. All formulas are read under the standing regularity and differential-operator conventions in Appendix D. In particular, $\mathbf{div}$ is row-divergence for tensor fields, $\Delta \boldsymbol{u}$ is the componentwise vector Laplacian, tensor norms are Frobenius norms when written, and η is dynamic viscosity while $\nu = \eta/\rho$ is kinematic viscosity as in Remark C.5.

F.1 Viscous-Term Simplification and Energy Context

For a Newtonian fluid with constant dynamic viscosity η and sufficiently smooth velocity field,

$$\mathbf{div}(2\eta \boldsymbol{D}(\boldsymbol{u})) = \eta \mathbf{div}(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^\top) = \eta \Delta \boldsymbol{u} + \eta \nabla(\mathrm{div} \boldsymbol{u}).$$

For incompressible flow, $\mathrm{div} \boldsymbol{u} = 0$, so

$$\mathbf{div}(2\eta \boldsymbol{D}(\boldsymbol{u})) = \eta \Delta \boldsymbol{u}, \qquad \rho^{-1} \mathbf{div}(2\eta \boldsymbol{D}(\boldsymbol{u})) = \nu \Delta \boldsymbol{u}, \qquad \nu = \eta/\rho.$$

This term is the momentum-diffusion contribution; it does not disappear in the incompressible Newtonian limit.

Classical local well-posedness results for Navier–Stokes depend on the domain, boundary conditions, forcing, initial-data space, and solution class. Under sufficiently smooth compatible data one obtains a unique strong solution on a maximal time interval. If the maximal interval is finite, continuation fails through blow-up of the norm controlled by the chosen well-posedness theory.

F.2 Cylindrical Coordinate Form

For (r, θ, z) with orthonormal basis $(\boldsymbol{e}_r, \boldsymbol{e}_\theta, \boldsymbol{e}_z)$,

$$x = r \cos \theta, \qquad y = r \sin \theta, \qquad \boldsymbol{u} = u_r \boldsymbol{e}_r + u_\theta \boldsymbol{e}_\theta + u_z \boldsymbol{e}_z.$$

The component formulas are local cylindrical-coordinate formulas on patches where $r > 0$. Axis regularity, wall traces, and boundary conditions require separate hypotheses and are not supplied by the coordinate notation alone. The differential operators used in cylindrical divided-by-density Navier–Stokes forms are

$$\begin{aligned}
(\boldsymbol{u}\cdot\nabla) &= u_r\partial_r + \frac{u_\theta}{r}\partial_\theta + u_z\partial_z,\\
\nabla p &= \partial_r p\,\boldsymbol{e}_r + \frac{1}{r}\partial_\theta p\,\boldsymbol{e}_\theta + \partial_z p\,\boldsymbol{e}_z,\\
\mathrm{div}\boldsymbol{u} &= \frac{1}{r}\partial_r(ru_r) + \frac{1}{r}\partial_\theta u_\theta + \partial_z u_z.
\end{aligned}$$

The rate-of-deformation components in the cylindrical orthonormal basis are

$$\begin{aligned}
&D_{rr} = \partial_r u_r, && D_{\theta\theta} = \frac{1}{r}\partial_\theta u_\theta + \frac{u_r}{r}, && D_{zz} = \partial_z u_z,\\
&D_{r\theta} = \frac{1}{2}\left(\partial_r u_\theta - \frac{u_\theta}{r} + \frac{1}{r}\partial_\theta u_r\right), && D_{rz} = \frac{1}{2}(\partial_r u_z + \partial_z u_r), && D_{\theta z} = \frac{1}{2}\left(\frac{1}{r}\partial_\theta u_z + \partial_z u_\theta\right).
\end{aligned}$$

Use the scalar shear-rate convention $\dot\gamma = \sqrt{2\boldsymbol{D}(\boldsymbol{u}):\boldsymbol{D}(\boldsymbol{u})}$. Then

$$\tau_{ij} = 2\eta_{\mathrm{eff}}(\dot\gamma)D_{ij}, \qquad i,j\in\{r,\theta,z\}.$$

When a source writes the law as a function of $D:D$ or $|D|_{\mathrm{F}}^2$, this report translates it through this convention, consistent with Remark C.5. Writing $\boldsymbol{f} = f_r\boldsymbol{e}_r + f_\theta\boldsymbol{e}_\theta + f_z\boldsymbol{e}_z$, the momentum components are

$$\begin{aligned}
\text{(radial)}\quad & \partial_t u_r + u_r\partial_r u_r + \frac{u_\theta}{r}\partial_\theta u_r + u_z\partial_z u_r - \frac{u_\theta^2}{r}\\
&= -\frac{1}{\rho}\partial_r p + \frac{1}{\rho}\left[\frac{1}{r}\partial_r(r\tau_{rr}) + \frac{1}{r}\partial_\theta\tau_{r\theta} + \partial_z\tau_{rz} - \frac{1}{r}\tau_{\theta\theta}\right] + f_r,\\
\text{(azimuthal)}\quad & \partial_t u_\theta + u_r\partial_r u_\theta + \frac{u_\theta}{r}\partial_\theta u_\theta + u_z\partial_z u_\theta + \frac{u_r u_\theta}{r}\\
&= -\frac{1}{\rho r}\partial_\theta p + \frac{1}{\rho}\left[\frac{1}{r}\partial_r(r\tau_{r\theta}) + \frac{1}{r}\partial_\theta\tau_{\theta\theta} + \partial_z\tau_{\theta z} + \frac{1}{r}\tau_{r\theta}\right] + f_\theta,\\
\text{(axial)}\quad & \partial_t u_z + u_r\partial_r u_z + \frac{u_\theta}{r}\partial_\theta u_z + u_z\partial_z u_z\\
&= -\frac{1}{\rho}\partial_z p + \frac{1}{\rho}\left[\frac{1}{r}\partial_r(r\tau_{rz}) + \frac{1}{r}\partial_\theta\tau_{\theta z} + \partial_z\tau_{zz}\right] + f_z.
\end{aligned}$$

The incompressibility condition is

$$\frac{1}{r}\partial_r(ru_r) + \frac{1}{r}\partial_\theta u_\theta + \partial_z u_z = 0.$$

G Numerical Methods Details

This appendix supplies the numerical-method detail behind Section 1.2.3 and the comparison boundary in Section 1.3. It is supporting method detail, not a claim of external-solver parity for the implemented solver. The continuous identities are read under Convention D.1; computed quantities are read through the following numerical-record convention.

Convention G.1 (Computed realization record)**.** A computed realization is interpreted only through the model, closure, discretization, grid, timestep policy, boundary-state rule, output operator, and diagnostic metadata recorded for that realization. Symbols such as D_t, D_z, numerical fluxes, residual arrays, and sampled norms denote recorded discrete objects unless a section explicitly switches back to a continuum identity.

The time interval, output rectangle, and norm notation follow Definition D.3 and Definition D.7; discrete residuals use the recorded D_t, D_z, grid, and quadrature weights rather than unnamed continuum derivatives.

G.1 Fixed-wall stationary-Stokes refinement record

The local three-dimensional refinement study is limited to a fixed-wall stationary-Stokes problem on the smooth idealized stenosis geometry. The public entrypoint `run_stationary_stokes_refinement` constructs generated Gridap tetrahedral meshes from the same C^∞ radius profile used by the reduced model, solves a pressure-drop-driven stationary Stokes problem, evaluates section-averaged velocity and pressure from the finite-element field, and compares those averages with the resistance/projection quantities used by the stationary-Stokes initializer. This keeps the initializer behavior unchanged: the study is an additional diagnostic record, not a replacement for the 1D initial-condition contract.

For a Newtonian fixed-wall Stokes field, the sampled stress and wall-traction diagnostics use

$$\begin{aligned} \boldsymbol{\sigma}(\boldsymbol{u},p) &= -p\mathbf{I} + \mu\left(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^\top\right), & \mu &= \rho\nu, \\ \mathbf{t}_w &= \boldsymbol{\sigma}\mathbf{n}, & \boldsymbol{\tau}_w &= \mathbf{t}_w - (\mathbf{t}_w\cdot\mathbf{n})\mathbf{n}. \end{aligned}$$

Here $\mathbf{n}$ is the analytic outward normal of the C^∞ wall profile. The reported means and maxima are fixed-wall fluid-stress diagnostics sampled from the finite-element solution. They are not structural displacement, wall velocity, structural stress, or resolved FSI interface-work outputs.

G.2 Elastic-potential balance-law form

The reduced state is $\mathbf{U} = (A, Q)^\top$. For a member of the $\mathcal{W}_{\text{elastic1D}}$ balance-law contract with prescribed momentum-flux correction α, the reference elastic-potential form is

$$\begin{aligned} \partial_t A + \partial_z Q &= 0, \\ \partial_t Q + \partial_z \left(\alpha \frac{Q^2}{A} + \Psi(A, z, t) \right) &= S_{\text{geom}}(A, z, t) - C_f(z, A, Q) \frac{Q}{A}. \end{aligned}$$

The elastic potential satisfies

$$\partial_A \Psi(A, z, t) = \frac{A}{\rho} \partial_A p_{\text{g}}(A, z, t),$$

and the geometry source is

$$S_{\text{geom}}(A, z, t) = \left. \partial_z \Psi(A, z, t) \right|_A - \frac{A}{\rho} \left. \partial_z p_{\text{g}}(A, z, t) \right|_A.$$

For the selected Canic–Koiter wall-law member, assuming p_{ext} is independent of A,

$$c^2(A, z) = \frac{A}{\rho} \partial_A p_{\text{g}}(A, z, t) = \frac{\beta(z)\sqrt{A}}{2\rho A_0(z)}.$$

Freeze z, the wall-closure data, and the source terms, and consider the homogeneous (A, Q) flux Jacobian associated with $(Q, \alpha Q^2/A + \Psi(A, z, t))^\top$. On a positive-area state $A > 0$, with flow rate Q, mean velocity Q/A, prescribed momentum-flux correction α, and $c^2(A, z) := (A/\rho)\partial_A p_{\text{g}}(A, z, t)$, its eigenvalues are

$$\lambda_\pm(A, Q; z) = \alpha \frac{Q}{A} \pm \sqrt{c^2(A, z) + \alpha(\alpha - 1) \left(\frac{Q}{A} \right)^2}.$$

For the displayed elastic wall law, $c^2 > 0$ when $\beta > 0$, $A > 0$, $A_0 > 0$, and $\rho > 0$. Strict hyperbolicity of the homogeneous elastic subsystem further requires the characteristic radicand

$$c^2(A, z) + \alpha(\alpha - 1) \left(\frac{Q}{A} \right)^2 > 0$$

to be positive on the reported state set. These speeds reduce to $Q/A \pm c(A, z)$ in the special $\alpha = 1$ reference case. The subcritical case used in Assumption 1.36 is $\lambda_- < 0 < \lambda_+$, so one characteristic enters through the inlet, one characteristic enters through the outlet, and one scalar boundary condition is imposed at each end of the vessel. The displayed derivatives of A_0, β, p_{g}, and Ψ are continuum notation for the model contract. A computed record may realize them through explicitly recorded finite-difference or source-split operators; changing that realization changes the computed map.

G.3 Semi-discrete solver operator

The implemented solver stores a radius-squared area coordinate in the variable `A`. In this subsection write that coordinate as $a = R^2$. The physical circular cross-sectional area is $A_{\mathrm{phys}} = \pi a$. This is a solver-specific convention; the continuum and classical 1D model sections keep A for physical cross-sectional area.

For N finite-volume cells, let

$$z_i = \left(i - \frac{1}{2}\right)\Delta z, \qquad \Delta z = \frac{L}{N}, \qquad u_h(t) = (a_1(t), \ldots, a_N(t), Q_1(t), \ldots, Q_N(t))^\top .$$

The semi-discrete system exposed by `semidiscretize` and `rhs!` is

$$\dot{u}_h = \mathcal{L}_h(u_h, t; \theta),$$

where θ denotes the recorded physical, geometry, grid, and solver parameters. Componentwise,

$$\begin{aligned}
\dot{a}_i &= -\frac{\widehat{F}^a_{i+1/2} - \widehat{F}^a_{i-1/2}}{\Delta z},\\
\dot{Q}_i &= -\frac{\widehat{F}^Q_{i+1/2} - \widehat{F}^Q_{i-1/2}}{\Delta z} + \widehat{S}_i.
\end{aligned}$$

All divisions by a use the positivity-regularized value of the radius-squared area coordinate.

With $K = Eh/(1-\sigma^2)$, Canic conservative-form reference radius $R_0^* = R_{\max}$, velocity-profile momentum factor $\alpha_{\mathcal{P}}$, and $\alpha_c(R_0') = -2(R_0')^2/35$, define $\iota_{\mathcal{M}} = 1$ for the `canic-extended-1d` forward-model descriptor and $\iota_{\mathcal{M}} = 0$ for `classical-1d-no-slip`. The semi-discrete full-state flux is

$$\mathbf{F}(a, Q, z) = \left(Q,\, (\alpha_{\mathcal{P}} + \iota_{\mathcal{M}}\alpha_c(R_0'(z)))\,\frac{Q^2}{a} + \frac{K}{3\rho R_{\max}^2} a^{3/2}\right)^\top .$$

At each interface,

$$\widehat{\mathbf{F}}_{i+1/2} = \frac{1}{2}\left[\mathbf{F}(\mathbf{U}^-_{i+1/2}, z_{i+1/2}) + \mathbf{F}(\mathbf{U}^+_{i+1/2}, z_{i+1/2})\right] - \frac{1}{2} s_{i+1/2}\left(\mathbf{U}^+_{i+1/2} - \mathbf{U}^-_{i+1/2}\right),$$

where $\mathbf{U} = (a, Q)^\top$ and the finite-volume realization uses neighboring first-order states plus ghost boundary states. The local numerical speed is

$$s_{i+1/2} = \max\left\{s_h(\mathbf{U}^-_{i+1/2}, z_{i+1/2}), s_h(\mathbf{U}^+_{i+1/2}, z_{i+1/2})\right\},$$

with

$$
\begin{aligned}
s_h(a,Q,z) = \max\left\{\left|\alpha_{\text{eff}}u-\chi\right|,\left|\alpha_{\text{eff}}u+\chi\right|\right\},\\
\alpha_{\text{eff}} &= \alpha_{\mathcal{P}}+\iota_{\mathcal{M}}\alpha_c(R_0'(z)), \qquad u=\frac{Q}{a},\\
\chi^2 &= (\alpha_{\text{eff}}u)^2-\alpha_{\text{eff}}u^2+\frac{K}{2\rho R_{\max}^2}\sqrt{a}.
\end{aligned}
$$

The source term uses the specified centered interior and one-sided boundary difference D_z^h on the cell averages. Let $g_{\mathcal{P}}$ denote the profile shear factor,

$$
\dot{\gamma}_{1D,i}=g_{\mathcal{P}}\frac{|Q_i|}{a_i\sqrt{a_i}}, \qquad \nu^{\mathcal{R}}_{\text{eff},i}=\frac{\eta^{\mathcal{R}}_{\text{eff}}(\dot{\gamma}_{1D,i})}{\rho},
$$

and $\alpha_c'(z)=-4R_0'(z)R_0''(z)/35$. The implemented source is

$$
\begin{aligned}
\widehat{S}_i &= -2\nu^{\mathcal{R}}_{\text{eff},i}g_{\mathcal{P}}\frac{Q_i}{a_i}+\frac{K}{\rho R_{\max}^2}a_iR_0'(z_i)-a_i\,\widehat{\partial_z p_2}_i+\iota_{\mathcal{M}}\frac{Q_i^2}{a_i}\alpha_c'(z_i),\\
\widehat{\partial_z p_2}_i &= \nu^{\mathcal{R}}_{\text{eff},i}g_{\mathcal{P}}\left[\frac{Q_iR_0''(z_i)}{a_iR_0(z_i)}-\frac{Q_i(R_0'(z_i))^2}{a_iR_0(z_i)^2}+\frac{(D_z^hQ)_iR_0'(z_i)}{a_iR_0(z_i)}-\frac{Q_i(D_z^ha)_iR_0'(z_i)}{a_i^2R_0(z_i)}\right].
\end{aligned}
$$

The $\widehat{\partial_z p_2}_i$ term is the discrete derivative of the Canic variable-radius viscous pressure correction. It is tied to the `canic-extended-1d` realization, while the elastic stiffness terms above come from the selected Canic–Koiter wall law. For the `newtonian` descriptor, this reduces to the constant value $\nu^{\mathcal{R}}_{\text{eff},i}=\nu$. For the non-Newtonian closures, the effective viscosity is evaluated with the shear-rate regularization and any declared interval projection before division by density. For the power-family profile, $g_{\mathcal{P}}=\gamma+2=\alpha_{\mathcal{P}}/(\alpha_{\mathcal{P}}-1)$. The flat profile is not evaluated through that singular limit; it is assigned $\alpha_{\mathcal{P}}=1$ and an explicit finite $g_{\mathcal{P}}$.

G.4 Time-integration realization

The native time integrator advances the semi-discrete operator with the declared third-order strong-stability-preserving Runge–Kutta map

$$
u_h^{n+1}=\Phi^{\text{SSPRK3}}_{\Delta t_n}\left(u_h^n;\mathcal{L}_h\right).
$$

In expanded form, using Π for the recorded positive-area projection on the a-components,

$$
\begin{aligned}
u^{(1)} &= \Pi\left(u^n+\Delta t_n\mathcal{L}_h(u^n,t^n)\right),\\
u^{(2)} &= \Pi\left(\frac{3}{4}u^n+\frac{1}{4}\left[u^{(1)}+\Delta t_n\mathcal{L}_h(u^{(1)},t^n+\Delta t_n)\right]\right),\\
u^{n+1} &= \Pi\left(\frac{1}{3}u^n+\frac{2}{3}\left[u^{(2)}+\Delta t_n\mathcal{L}_h(u^{(2)},t^n+\Delta t_n)\right]\right).
\end{aligned}
$$

The native step size is

$$\Delta t_n = \min\left\{\Delta t_{\max}, \frac{\mathrm{CFL}\,\Delta z}{\max_i s_h(a_i^n, Q_i^n, z_i)}, T_{\mathrm{final}} - t^n\right\}.$$

The SciML realization constructs an `ODEProblem` for the same $\dot{u}_h = \mathcal{L}_h(u_h, t; \theta)$ and applies the selected OrdinaryDiffEq policy recorded in the solve specification. Thus comparisons between these realizations change the time integrator, not the semi-discrete spatial operator.

G.5 Boundary-state realization

The finite-volume boundary realization imposes one incoming boundary condition at each boundary in the subcritical regime using an $\alpha = 1$, fixed-area characteristic approximation. This is the implemented boundary-state rule for the variable-α_{eff} solver, not an exact invariant derivation for the full variable-radius balance law. Let

$$c_0 = \left(\frac{K}{2\rho(R_0^*)^2}\right)^{1/2} = \left(\frac{K}{2\rho R_{\max}^2}\right)^{1/2}.$$

At the inlet, the incoming flow is $Q_{\mathrm{in}} = R_0(0)^2 \bar{u}_{\mathrm{in}}$. The outgoing invariant from the first interior cell is

$$w_- = \frac{Q_1}{a_1} - 4c_0 a_1^{1/4},$$

and the ghost coordinate a_{in} solves

$$\frac{Q_{\mathrm{in}}}{a_{\mathrm{in}}} - 4c_0 a_{\mathrm{in}}^{1/4} = w_-.$$

At the outlet, the radius-squared coordinate is fixed to $a_{\mathrm{out}} = R_0(L)^2$. The outgoing invariant from the last interior cell is

$$w_+ = \frac{Q_N}{a_N} + 4c_0 a_N^{1/4},$$

and the ghost flow is

$$Q_{\mathrm{out}} = a_{\mathrm{out}} w_+ - 4c_0 a_{\mathrm{out}}^{5/4}.$$

The boundary update therefore constrains the incoming information and leaves the outgoing information tied to the interior state. Boundary-quality plots and reflected-wave sidecars are diagnostics until explicit thresholds are admitted.

G.6 Residual diagnostics

Stored residuals are post-processing diagnostics on sampled fields, not proof of the continuous PDE residual. The symbols D_t and D_z in this subsection denote the recorded temporal and spatial finite-difference operators on the saved grid, not weak derivatives. For a stored field sequence, the semi-discrete defect is read as

$$\mathcal{E}_h^n = D_t u_h^n - \mathcal{L}_h(u_h^n, t^n; \theta).$$

Componentwise, this gives

$$\mathcal{E}_i^a(t^n) = D_t a_i^n - \mathcal{L}_{h,i}^a(u_h^n, t^n; \theta), \qquad \mathcal{E}_i^Q(t^n) = D_t Q_i^n - \mathcal{L}_{h,i}^Q(u_h^n, t^n; \theta).$$

The E1 diagnostic summary reports these residuals on reviewed interior cells with recorded finite-field, positivity, CFL, and pressure-drop sign checks. The numbers support the dataset contract and local consistency reading; they do not establish a convergence theorem or external parity. Any residual norm or error norm reported from these arrays is a sampled norm tied to the numerical record, rather than an unqualified continuum $L^p(K)$ norm.

H Code Availability and Build Statement

This report is built from the LaTeX sources rooted at `final-report.tex`. The compiled PDF should be treated as the rendered form of those sources. The report package used for this build is the repository root containing `final-report.tex`. Because no public remote or archival release is declared here, this appendix does not claim a public repository or archival URL.

The Julia simulation code is packaged as `CanicExtended1D` in the root Julia project. The tracked source inventory is:

Tracked source	**Role in the report package**
`Project.toml, Manifest.toml`	Julia package identity, dependency declarations, compatibility bounds, and resolved dependency versions.
`src/CanicExtended1D.jl`	Package entry point and public export surface for the solver, studies, resolved-3D comparison, and report figure export helpers.
`src/CanicExtended1D/*.jl`	Solver components: model descriptors, geometry, state layout, finite-volume and DG operators, native and SciML backends, Stokes initial condition, OpenBF-style adapter, studies, refinement, resolved-3D diagnostics, output writers, package benchmarks, and geometry-export helpers.
`simulations/run_canic_extended_1d.jl, simulations/canic_extended_1d_stenosis.jl`	Thin command-line wrappers around `CanicExtended1D.run_cli`.
`simulations/run_package_benchmark.jl`	Thin command-line wrapper around `CanicExtended1D.run_package_benchmark`; writes the benchmark manifest and CSV schemas used in Section 2.1.
`simulations/export_stenosis_geometry_figures.jl`	Thin wrapper around `CanicExtended1D.export_stenosis_geometry_figures`.
`python/src/research_hemodynamics/*.py`	Python Typer CLI and native/Torch/SciPy/SciML adapter surface used for descriptor smoke runs and CPU/Torch-MPS comparison rows.
`scripts/render_package_benchmark_figures.py`	Python renderer for package-benchmark CSVs, report figures, and compact LaTeX table fragments.
`test/runtests.jl, test/test_*.jl`	Julia test orchestrator and feature-specific regression tests for model closures, backends, comparison diagnostics, CLI parsing, studies, benchmarks, and exports.
`scripts/julia-release`	Repository launcher that selects Julia 1.12 or newer and binds the root project environment.

The standard package import surface used by report regeneration commands is:

```julia
using CanicExtended1D

r = run_available_resolved3d_comparison(
    output_dir="simulations/output/3d_comparison/full_t1_native_nx400",
    overwrite=true,
)
publish_resolved3d_report_assets(r; overwrite=true)
```

The package-benchmark suite is regenerated separately from the tracked PDF. A smoke run writes deterministic schema checks; an overnight run expands the same schemas to the full benchmark matrix:

```
./scripts/julia-release simulations/run_package_benchmark.jl \
  --profile overnight \
  --output-dir simulations/output/package_benchmark/<run_id> \
  --overwrite \
  --include-python \
  --include-resolved3d \
  --publish-report-assets

pipenv run python scripts/render_package_benchmark_figures.py \
  --benchmark-dir simulations/output/package_benchmark/<run_id>
```

The tracked package-benchmark assets used in Section 2.1 are copied under `figures/static/static/data/package-benchmark/` and rendered under `figures/static/static/rendered/`. The tracked manifest records an `overnight` profile generated at `2026-06-18T20:20:04Z`. Its source output directory was the `overnight-20260618-quadrature/` benchmark run; Python stages and resolved-3D stages were enabled, and the completed run was copied into the tracked report assets. The recorded runtime context was Julia `1.12.6`, Python `3.13.12`, Torch `2.12.0`, MPS available, SciPy unavailable, and generation commit `24d311c0ddd169f943169b05736d439d9b877b57`. That commit identifies the benchmark/data-generation base state, not by itself the complete integrated manuscript state after the final report edits. The integrated report state is identified by the tracked source tree containing this appendix together with the file hashes below. The published benchmark CSVs contain 252 status rows across the eight report stages, all marked `ok`; this is an internal status summary, not an external validation claim.

The tracked package-benchmark data and table fragment have the following SHA-256 hashes:

Tracked package-benchmark file	SHA-256
figures/static/static/data/package-benchmark/manifest.json	dde2101fcea3d9cc896847682835356f 30a5d9c1d1aaa486fc7587a6f03bee50
figures/static/static/data/package-benchmark/case_results.csv	0843a3e3fa986e0ec1227bedab1ff8f 616e3d3652b9e5e609c64a171e6d124fe
figures/static/static/data/package-benchmark/refinement.csv	de20c481fbbfe4a5930d996887a55a2c 3fe3734de40422ff9fc83f4f4fc59e7e
figures/static/static/data/package-benchmark/backend_parity.csv	1a52b83cebb750cca68ac68f1a508dda 11b1c649ed62f71fe239582ee4ddf671
figures/static/static/data/package-benchmark/stokes_ic.csv	2379d100d1b01943b1afcd79718f298 10bad0fec9762796c72d78158963d6adb
figures/static/static/data/package-benchmark/rheology_profile.csv	8c83c6119d54dd3465eb8c3ec1e96897 2a981bd41afdbe34861ea4151c4fb62a
figures/static/static/data/package-benchmark/boundary_openbf.csv	90fcac4d39e1e0b705837597ff15b929 cb165a84fc56853a70839d53914c993d
figures/static/static/data/package-benchmark/resolved3d.csv	96443ae2dd54f9a0132ac7979209b256 0353195ee7ebb87d07a4778b66d1a53a
figures/static/static/data/package-benchmark/python_mps.csv	a50dfc626f2a59f1ace544337e4c1d09 0168b09e2b502f92abf3b1b4cf3177cb
figures/static/static/tables/package-benchmark/package-benchmark-summary.tex	9649ee00c26921196d628a2c9f17f7d8 14e4390ad194da8f399482aafa617948

The tracked package-benchmark rendered figures used by the PDF have the following SHA-256 hashes:

Tracked rendered file	SHA-256
figures/static/static/rendered/package-benchmark-convergence.pdf	08bbb0dd2594681ccd8f2dda98a2c802 96cc53af4ddc94c5b60f2c72c21a33a2
figures/static/static/rendered/package-benchmark-backend-parity.pdf	281d851f2d66f031fce4172da88d333d 0bfefbccc80bf9c183a147f0b59baa06
figures/static/static/rendered/package-benchmark-rheology-profile.pdf	aac13d7365ea098030aeafaeb9f3bb0f 7a9bdec4fda4e46d682ad6d243e70c6c
figures/static/static/rendered/package-benchmark-resolved3d.pdf	9b6282153ee2827007df74a1d394f1ec 82e7b4d8ab93b6c176d2a102c4af4cab
figures/static/static/rendered/package-benchmark-python-mps.pdf	2a84d08be281de3a8c81e2710065bfc1 1400be8a258a2feb401452d9a77a8817

The Section 2 comparison artifact is the non-archived directory `simulations/output/3d_comparison/full_t1_native_nx400/`. The command shape for regenerating the comparison, when the raw 3D inputs are restored, is:

```
./scripts/julia-release -e 'using CanicExtended1D; r=run_available_resolved3d_comparison(
    output_dir="simulations/output/3d_comparison/full_t1_native_nx400", overwrite=true);
    publish_resolved3d_report_assets(r; overwrite=true)'
```

The raw upstream XDMF/HDF5 inputs used for Figure 5 and Section 2 were restored locally from `https://github.com/qcutexu/Extended-1D-AQ-system`, commit `056a9da2b36b480691f18025d242d2c00f6e7180`, under `case3_all_3d_results/`. The local copies under `simulations/data/3d/canic_case3/` are ignored by git and are not archived in this report package. Rerunning the raw XDMF/HDF5 extraction therefore requires those local inputs or the upstream repository. The observed local raw velocity-input hashes are:

Ignored local input	SHA-256
simulations/data/3d/canic_case3/77/velocity.xdmf	8cbb61e8d6a8b9fbfe9662bea7544eb8 9ef58390cdcc93a5d0f68fbb19c110b7
simulations/data/3d/canic_case3/77/velocity.h5	315157b3c9a9a92d1e99b26c3e9c393f 22303a676b3b24f52287d0b8cd22d168
simulations/data/3d/canic_case3/60/velocity.xdmf	765adb3f603b2306144dfb9be6ffec5573 23686867172ea9cc7b6e3f246284e0
simulations/data/3d/canic_case3/60/velocity.h5	1dee423991e35ced10a051e20b0598ee 0cbea3d59521b0fc3cb490b49bb4fe41

The tracked static data copied into the PDF has the following SHA-256 hashes:

Tracked file	SHA-256
figures/static/static/data/stenosis-comparison/ section-quadrature.dat	92a6df1b035373abf9cf4fdfd2df7254 fb52c7d46378ff3ce89345d9ec34ee2c
figures/static/static/data/stenosis-comparison/ radial-quadrature-C23.dat	27f2f12747bfb1eb87e62cac736c3df1 1b94b0a4f02b3b3ebf157e6b72435bcd
figures/static/static/data/stenosis-comparison/ radial-quadrature-C40.dat	4e0a6527b87f5b395e96cf379346c288 9175e8e9181a7d3f4b2423b57df024f0
figures/static/static/data/stenosis-comparison/ node-slab-sensitivity.csv	23014b8c5342fce4bfb1e7878fff815d b988082b0ba344fbcbc688042cd5c167
figures/static/static/data/stenosis-comparison/area-audit.dat	98e1fbb5b06ebe8cd928784e8c48fdc3 9cc07b0dde228c58edaf82107a738696
figures/static/static/data/stenosis-geometry/resolved_ velocity_nodes_manifest.csv	372796413873c09581ced890f8de657e f39e9ff69eabca8ce46b51b0cf758d3f
figures/static/static/data/stenosis-geometry/resolved_ velocity_nodes_case77_sev23.csv	0c0451dcee29216d967cecb11d36f518 a95c469fbcf18a1b472d7b9f01d211d0
figures/static/static/data/stenosis-geometry/resolved_ velocity_nodes_case60_sev40.csv	eb4b2d5149128740bfbb89ff219e92a47 d0535f3a7fd45078732b837619f68d1
figures/static/static/data/stenosis-geometry/resolved_ envelope_manifest.csv	d8236b5549851ff97eb9d9628b2bfbc6 60f0e594738b634509d0538476ef25a7
figures/static/static/data/stenosis-geometry/resolved_ envelope_case77_sev23.csv	7ffbbdbf52e4cab5c27e7aebc54bc47d 8aa0e7e2ab2b7058c61da8f06bb40192
figures/static/static/data/stenosis-geometry/resolved_ envelope_case60_sev40.csv	e661114841a0b3f8abccb0dc244e2d4 b4409291c8f02dcd5a4da2398892a7c1d
figures/static/static/rendered/resolved-3d-flow-field.pdf	dbee4f2bc37c980c8ce99f172e26edbf 90eaf8cb1b23e1fb787a77298a4e8aca

The non-archived comparison artifact observed for this build contains the following output hashes; these files are provenance evidence, not tracked archive contents:

Generated artifact file	SHA-256
comparison_summary.csv	c582ee73a73b8c988a185bf55e86943c a7a459e9ad74f6826ca83bddafb165a2
section_mean_case77_sev23.csv	641a7ca341f9e6a6d1eecfd13ab94a8 4fcd77d5eb79ff4ebec63efa47605c39e
section_mean_case60_sev40.csv	807c548d90cbb21fbecc333cfc4f185e a60e05e0b317a4d34d480026d897afa2
radial_profile_case77_sev23.csv	781018d4772472479a3b4f76ed61e10e 7bdefa7786c2daba263e2a130fb2b56a
radial_profile_case60_sev40.csv	d29c698ec8c2c35ffa39f51fa24e6b0d 96a60049fae20a7a65722050e7ca9ad4
node_slab_sensitivity.csv	23014b8c5342fce4bfb1e7878fff815d b988082b0ba344fbcbc688042cd5c167
section_quadrature_overlay.svg	2b465f2593106cac942c5f3733cde046 23dbe51b6c2f4ebfa81486d6b59dab94

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